

A Conformal-Anomaly Resolution of the Cosmological Constant Problem

The Vacuum Energy, Its Value, the Coincidence, and a Parameter-Free Curvature Fingerprint — with a Complete Account of What Is Proven and What Is Conditional

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Abstract

The cosmological constant problem has three faces: the *magnitude* problem (why the observed vacuum energy is $\sim 10^{-120}$ in Planck units rather than of order unity), the *value* problem (what fixes the residual scale once the radiative contributions are removed), and the *coincidence* problem (why the dark-energy and matter densities are comparable now). This paper assembles a single, internally consistent framework that addresses all three under explicit, stated assumptions — notably the physical identifications A5 and P5 made precise below — and, as a by-product, yields one sharp falsifiable prediction. The magnitude problem is handled by Kaloper–Padilla vacuum-energy sequestering, which removes the radiative vacuum energy from the gravitating sector. The value problem is addressed by a self-consistency result (Proposition 2): the residual cosmological constant is a fixed point of the four-volume averaging map $\Lambda = \frac{1}{4}\langle R \rangle_{\text{ren}}$, equivalent to the trace-zero condition $\langle T \rangle_{\text{ren}} = 0$, and is therefore *not* a free parameter; the uniqueness of that fixed point rests on a Banach contraction whose constant is so far established numerically. The coincidence problem is addressed by structure-formation (halo-weighted) averaging following Lombriser, returning $\Omega_\Lambda \approx 0.704$ conditional on an anthropic-like prior; the framework’s own budget ladder instead points to $\Omega_\Lambda = 4a_\gamma = 0.6889$, coinciding with the robust Planck+BAO value ($\Omega_m = 0.3111$) to four decimals — an empirical coincidence, not a proof: the absolute value rests on the open holographic-saturation conjecture P5 (Section 16.1). Threading through all three is a relation — parameter-free as a ratio of two scale-free invariants, though its identification with the *observable* curvature rests on the physical postulate A5 (below) — between the residual cosmological constant and the spatial curvature,

$$\frac{|\Omega_K|}{\Omega_\Lambda} = \frac{a_\gamma}{8\pi^2} = \frac{31}{1440\pi^2} \approx 2.181 \times 10^{-3}, \quad \mathcal{L} \equiv \frac{\Omega_\Lambda}{|\Omega_K|} = \frac{8\pi^2}{a_\gamma} \approx 458.4,$$

whose two inputs — the photon type-A conformal anomaly coefficient $a_\gamma = 31/180$ and the Q -curvature charge $C_Q = \int_{D^4} Q_4 = 8\pi^2$ of the hemisphere — contain no scale, no Newton constant, and no Planck mass. We give the rigorous spectral backbone in full: twelve proven spectral results (R1–R12), the Burghlelea–Friedlander–Kappeler gluing identity, the cap-variational saddle and its stability in every sector, the higher-loop convergence bound, and an independent recomputation (M1) with exact symbolic and numerical methods. Every step carries an explicit status — *proven*, *literature-supported*, *calibrated*, or *open*. After decomposition, the framework reduces to a single genuinely chosen element — the data-anchored reference state A5c — plus the embedding into the full infinite-dimensional no-boundary path integral (the two-dimensional cap-plus-scalar embedding $B1'$ being complete, $n_{\text{complex}} = 0$ by polynomial reduction); its topological core (the ratio A5a, the self-consistency fixed point A2, cap dominance B1 on the reduced action) is theorem-level, and the open sign is forced by the positive photon anomaly coefficient $a_\gamma > 0$ *within the Hartle–Hawking branch* (the branch selection itself a postulate). The prediction is sharply falsifiable: a robust Ω_K outside $\pm 3\sigma$ of $a_\gamma\Omega_\Lambda/(8\pi^2)$, or a robust detection of a closed universe, would refute the framework as stated.

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1 The cosmological constant problem and what a resolution must deliver

The cosmological constant Λ is the sharpest quantitative conflict between quantum field theory and general relativity: naive estimates of the quantum-vacuum zero-point energy exceed the observed dark-energy density by some 120 orders of magnitude. The problem is not one question but three, and any framework claiming a *resolution* must say something about each.

(i) **The magnitude problem.** Why is the gravitating vacuum energy $\rho_\Lambda \sim 10^{-120} M_{\text{Pl}}^4$ rather than $O(M_{\text{Pl}}^4)$? Every massive field contributes a quartically divergent vacuum energy; the cancellation to 120 digits is the heart of the old cosmological constant problem.

(ii) **The value problem.** Suppose the radiative contributions are removed by some mechanism. What then fixes the residual, nonzero scale? A mechanism that sets Λ to zero is no better than one that leaves it free: the observed value is small but not zero.

(iii) **The coincidence problem.** Why are $\Omega_\Lambda \approx 0.69$ and $\Omega_m \approx 0.31$ of the same order *now*, when their ratio scales as a^3 and has therefore been wildly different at almost every other epoch?

The strategy of this paper. We address (i) by Kaloper–Padilla sequestering (Section 8), which removes the radiative vacuum energy from the source of Einstein’s equations. We address (ii) by a self-consistency result (Section 10): the residual Λ is a fixed point of a four-volume

averaging map, hence dynamically determined rather than free. We address (iii) by structure-formation averaging (Section 16), which returns the observed $\Omega_\Lambda \approx 0.704$ in Lombriser’s halo-weighted prescription, conditional on an anthropic-like prior; the framework’s own — conjectural — budget ladder instead points to $\Omega_\Lambda = 4a_\gamma = 0.6889$, coinciding with the robust Planck+BAO datum to four decimals — an empirical coincidence, not a proof (Section 16.1). The framework’s sharpest — and only sharply falsifiable — output is a relation, parameter-free as a ratio of two scale-free invariants, between the residual cosmological constant and the present spatial curvature,

$$\boxed{\frac{|\Omega_K|}{\Omega_\Lambda} = \frac{a_\gamma}{8\pi^2} = \frac{31}{1440\pi^2} \approx 2.181 \times 10^{-3}} \quad (1)$$

with $a_\gamma = 31/180$ the photon type-A conformal anomaly coefficient and $8\pi^2$ the Q -curvature charge of the four-hemisphere $D^4 \subset S^4$. In the present epoch the ladder-consistent budget $\Omega_m = 0.3096$ gives, by the Friedmann sum rule, $|\Omega_K| \approx 1.50 \times 10^{-3}$ and $\Omega_\Lambda = 4a_\gamma = 0.6889$ (Section 16.1).

What is claimed, and what is not. The account rests on three disclaimers, stated plainly at the outset.

1. The relation (1) is a statement about a *ratio*. The *absolute* dark-energy density is not a *sharp* parameter-free output of the topology: the KP self-consistency route fixes Λ *structurally* (not free, sign negative, the bulk vacuum cancels) and delivers a *universal* fixed point Λ^* (independent of φ_0 and Ω_m), but its value is amplitude-set and the absolute scale is left open because φ_0 stays free (Section 10; the earlier $\langle R \rangle/4$ sharpness proxy was a floored-geometry artefact). The sharp value is then either supplied by structure-formation (halo) averaging under one additional assumption (Section 16), bracketed prior-free by the causal closure (Section 16.2), or — conjecturally — fixed topologically by the geometric budget ladder $\Omega_\Lambda = 4a_\gamma$ (Section 16.1), whose lift prefactor is computed ($= A/G$, the Bekenstein–Hawking horizon degree-of-freedom count) and whose residual — the holographic Cohen–Kaplan–Nelson saturation P5 — is localized but not proven. This residual is confined to the *absolute* value: the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ is scale-invariant, so the falsifiable Euclid prediction stands independently of it.
2. The Euclidean conformal and topological core is theorem-level — including the topological ratio A5a (Cauchy + Chern–Weil + APS), the KP self-consistency fixed point A2 (M5 Banach contraction), and cap-saddle dominance B1 on the reduced action ($n_{\text{complex}} = 0$, analytic). The bridge to the observed Lorentzian Friedmann curvature rests on a single remaining physical postulate, the identification A5b/A5c (Section 11), decided by observation; the two-dimensional cap-plus-scalar embedding $B1'$ is itself now complete ($n_{\text{complex}} = 0$ by a degree-8 polynomial reduction, Section 18), and that embedding into the full infinite-dimensional no-boundary path integral is assembled semiclassically (Section 18): the inhomogeneous one-loop determinant carries the anomaly ($\zeta(0) = -4a_\gamma$) and the Type-A coefficient is Adler–Bardeen one-loop exact, and the prediction $a_\gamma/(8\pi^2)$ is all-orders stable as a ratio of $\hbar$ -independent invariants (Section 18); the only residue is the absolute Borel summability of the matter series, decoupled from the prediction.
3. The *sign* (open versus closed) is forced by the positive photon anomaly coefficient $a_\gamma > 0$: the cap action has a unique real saddle at $\delta\theta^* > 0$ whose location is independent of the Hartle–Hawking/Vilenkin weighting (Section 17, Section 18). The separate question of Lorentzian fluctuation stability (Feldbrugge–Lehners–Turok) is a foundational input of all no-boundary cosmology, not specific to this framework.

A historical analogy. A clean, parameter-free relation justified at the level of a topological/anomaly identity, with a physically motivated but not yet rigorously derived bridge to the observable, has a precedent. Bekenstein’s 1972 conjecture that black-hole entropy is proportional

to horizon area [1] was a thermodynamic argument; its microscopic derivation arrived twenty-three years later with Strominger and Vafa [2]. The present framework is offered in the same spirit.

Outline. Sections 2–5 assemble the three rigorous ingredients: the photon anomaly coefficient a_γ , the Q -curvature charge C_Q , and the spectral backbone (twelve results R1–R12 and the BFK identity). Sections 8 and 10 treat the magnitude and value problems (Kaloper–Padilla sequestering and the self-consistency result). Section 11 derives relation (1) — parameter-free given the identification A5 — in eight steps and isolates the residual reference choice A5c; Section 12 gives the cap-variational mechanism, its stability, the boundary refinement, and the prediction hierarchy; Section 14 establishes uniqueness, within stated criteria, against 26 alternative mechanisms. Section 16 addresses the coincidence; Section 17 establishes the sign within the Hartle–Hawking branch; Section 19 gives the physical interpretation; Section 20 the higher-order corrections; Section 21 the independent computational verification (M1); Section 23 the observational status; and Section 25 the consolidated status ledger and falsification criteria.

2 The conformal anomaly and the photon coefficient

2.1 Trace anomaly and the type-A coefficient

A classically conformally invariant field has a traceless stress tensor, $T^\mu{}_\mu = 0$. Quantization breaks this; in four curved dimensions the renormalized trace acquires the universal form [3, 4]

$$\langle T^\mu{}_\mu \rangle = \frac{1}{16\pi^2} \left(c C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} - a E_4 + b \square R \right), \quad (2)$$

where C is the Weyl tensor, $E_4 = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 4R_{\mu\nu} R^{\mu\nu} + R^2$ is the Gauss–Bonnet (Euler) density, and a, c are the type-A and type-B central charges. The $b \square R$ term is scheme-dependent and integrates to zero on a closed manifold. The Deser–Schwimmer theorem [3] establishes that this classification is complete and geometric.

2.2 Why only the photon survives in the infrared

On the present-day de Sitter-like background the Hubble scale H_0 lies far below the mass of every massive field. By the Komargodski–Schwimmer monotonicity of the a -coefficient under renormalization-group flow [5], massive fields decouple from the infrared conformal sector; only the genuinely massless gauge field — the photon — still contributes to the type-A anomaly at cosmological scales. This is the content of Axiom A3 (photon-uniqueness), placed on a quantitative footing in Section 14 by the exhaustion of 26 standard-Lagrangian alternatives.

2.3 The value $a_\gamma = 31/180$

The photon type-A coefficient is a textbook result of the DeWitt–Schwinger heat-kernel expansion for a Maxwell field [6, 7, 4]:

$$a_\gamma = \frac{31}{180}. \quad (3)$$

We *derive* Eq. (3) from the Gilkey heat-kernel coefficient a_4 in Section 3: the Maxwell combination (Hodge Laplacian on 1-forms minus two Faddeev–Popov ghosts) gives $\zeta(0; \text{Maxwell}; S^4) = -31/45$, hence $a_\gamma = 31/180$, with the conformal-scalar value $1/360$ serving as an independent check. The same coefficient governs the closed-manifold zeta value $\zeta(0; \text{Maxwell}; S^4) = -4a_\gamma = -31/45$ (Spectral Result R1 below), since the Weyl term vanishes on the conformally flat S^4 and $\int_{S^4} E_4 = 64\pi^2$ gives $\int_{S^4} \langle T^\mu{}_\mu \rangle = 4a_\gamma$.

3 Heat-kernel coefficients

The DeWitt–Schwinger expansion of a second-order elliptic operator $D = -\square + E$ has $K(x, x; s) = (4\pi s)^{-n/2}[\mathbf{1} + s a_1 + s^2 a_2 + \dots]$. In $n = 4$ the anomaly-relevant coefficient is

$$a_2 = \frac{1}{180}(R_{\mu\nu\rho\sigma}^2 - R_{\mu\nu}^2)\mathbf{1} + \frac{1}{72}R^2\mathbf{1} - \frac{1}{30}\square R\mathbf{1} + \frac{1}{6}\square(\frac{R}{6}\mathbf{1} - E) + \frac{1}{2}(\frac{R}{6}\mathbf{1} - E)^2 + \frac{1}{12}W_{\mu\nu}W^{\mu\nu}.$$

For the Maxwell vector ($E^\nu{}_\mu = -R^\nu{}_\mu$, $\text{tr } \mathbf{1} = 4$) plus the Faddeev–Popov ghost ($E = 0$, contributing with a -2), the trace gives $\text{tr } a_2 = \frac{1}{16\pi^2}[a_\gamma E_4 - c_\gamma W^2 + b\square R]$ with $a_\gamma = 31/180$, $c_\gamma = 1/10$, and b a removable counterterm.

Explicit evaluation of a_γ on S^4 . The coefficient a_γ is not assumed; it follows from Gilkey’s integrated a_4 . Dropping total derivatives on a closed manifold,

$$\int a_4 = \frac{1}{(4\pi)^2 360} \int \sqrt{g} \text{tr} \left\{ 60 R E + 180 E^2 + 30 \Omega_{\mu\nu} \Omega^{\mu\nu} + \mathbf{1} (5R^2 - 2R_{\mu\nu}^2 + 2R_{\mu\nu\rho\sigma}^2) \right\},$$

with $\Omega_{\mu\nu}$ the bundle curvature of D . On S^4 one has $R_{\mu\nu}^2 = R^2/4$, $R_{\mu\nu\rho\sigma}^2 = R^2/6$, $W^2 = 0$, and $\int E_4 = 64\pi^2$, so only the type-A (Euler) part survives and the integrated conformal anomaly equals $\int \langle T^\mu{}_\mu \rangle = -4a_\gamma$ for a single field. Three operators are needed:

- *Conformal scalar* ($D = -\square + \frac{R}{6}$, $E = -\frac{R}{6}$, $\dim 1$, $\Omega = 0$): the bracket reduces to $-\frac{1}{6}R^2$, giving $\int \langle T \rangle = -\frac{1}{90}$ and $a_{\text{scalar}} = \frac{1}{360}$ — the textbook value, a check on the whole computation.
- *Hodge Laplacian on 1-forms* ($E = -\text{Ric}$, $\text{tr } E = -R$, $\text{tr } E^2 = R^2/4$, $\dim 4$, $\text{tr } \Omega_{\mu\nu} \Omega^{\mu\nu} = -R_{\mu\nu\rho\sigma}^2 = -R^2/6$): $\int \langle T \rangle_{\Delta_1} = -\frac{2}{45}$.
- *Minimal scalar ghost* ($E = 0$, $\dim 1$): $\int \langle T \rangle_{\Delta_0} = \frac{29}{90}$.

The photon partition function is $\ln Z = -\frac{1}{2} \ln \det \Delta_1 + \ln \det \Delta_0^{\text{ghost}}$, so its anomaly is the combination $\Delta_1 - 2 \Delta_0$:

$$\zeta(0; \text{Maxwell}; S^4) = \int \langle T \rangle_{\Delta_1} - 2 \int \langle T \rangle_{\Delta_0} = -\frac{2}{45} - 2 \cdot \frac{29}{90} = -\frac{31}{45}, \quad a_\gamma = -\frac{1}{4} \left(-\frac{31}{45} \right) = \boxed{\frac{31}{180}}.$$

This reproduces Spectral Result R1 and fixes the central input of the framework from the heat-kernel coefficient alone, so a_γ is derived here rather than adopted as an external citation.

4 The Q -curvature charge of the hemisphere

4.1 The Q -curvature and the Paneitz operator

Branson’s fourth-order Q -curvature in four dimensions is

$$Q_4 = -\frac{1}{6}\square R - \frac{1}{2}R_{\mu\nu}R^{\mu\nu} + \frac{1}{6}R^2, \quad (4)$$

the natural curvature scalar transforming covariantly under the Paneitz operator P_4 via $P_4 \sigma + Q_4 = Q'_4 e^{4\sigma}$ under a Weyl rescaling $g \rightarrow e^{2\sigma} g$. The Chang–Yang theory of extremal functional determinants [8, 9] singles out $\int Q_4$ as the conformally natural total curvature: only $\int Q_4$ (not $\int E_4$) transforms covariantly under P_4 , which is why Q_4 , and hence the charge C_Q , is the correct denominator of the topological quotient (1). On any closed Einstein four-manifold $\int Q_4 = \frac{1}{4} \int E_4$ exactly (Spectral Result R12).

4.2 The hemisphere value $C_Q = 8\pi^2$, with an elementary proof

Lemma 1 (Hemisphere Q -charge). *On the round unit hemisphere $D^4 = \{0 \leq \theta \leq \pi/2\} \subset S^4$, $C_Q \equiv \int_{D^4} Q_4 \sqrt{g} d^4 x = 8\pi^2$.*

Proof. On the unit S^4 (an Einstein manifold) $R = 12$, $R_{\mu\nu} = 3g_{\mu\nu}$, $\square R = 0$, hence $Q_4 = \frac{1}{6}(R^2 - 3R_{\mu\nu}R^{\mu\nu}) = \frac{1}{6}(144 - 108) = 6$. With the metric volume element $\sqrt{g} = \sin^3 \theta$ and $\text{Vol}(S^3) = 2\pi^2$,

$$\int_0^{\pi/2} \sin^3 \theta d\theta = \left[-\cos \theta + \frac{1}{3} \cos^3 \theta \right]_0^{\pi/2} = \frac{2}{3}, \quad \int_{D^4} Q_4 \sqrt{g} d^4x = 6 \cdot 2\pi^2 \cdot \frac{2}{3} = 8\pi^2. \quad \square$$

This elementary three-step evaluation has been confirmed by exact symbolic computation (Section 21, M1-Q). The constant $8\pi^2$ is the exact, scale-free Q -curvature charge of the hemisphere; it is a *proven* input. The cap-saddle shift $\delta\theta^* \approx 0.164$ introduced in Section 12 is a position in the wave-function argument, *not* a deformation of this integration domain; the charge is always evaluated on the standard hemisphere, so the leading prediction $|\Omega_K|/\Omega_\Lambda = a_\gamma/(8\pi^2)$ carries no $\delta\theta^*$ correction. The cap-saddle enters only at the refinement level L1, where it replaces the numerator a_γ while the denominator $8\pi^2$ stays fixed.

5 Spectral foundations: BFK decomposition and the twelve results

The bridge between the closed S^4 and its hemisphere D^4 is the Burghlea–Friedlander–Kappeler (BFK) gluing formula for functional determinants [10], which relates the determinant on a closed manifold to those on two pieces glued along a hypersurface plus the Dirichlet-to-Neumann (DtN) operator Λ_{S^3} on the equatorial S^3 . For the zeta-regularized value at zero,

$$\zeta(0; \Delta_{S^4}) = 2\zeta(0; \Delta_{D^4}^{\text{Dir}}) - \zeta(0; \Lambda_{S^3}). \quad (5)$$

5.1 The twelve spectral results R1–R12

The framework rests on twelve proven spectral results, all independently verified (several to high numerical precision). They are the rigorous backbone of the construction; we state them compactly.

- R1.** $\zeta(0; \text{Maxwell}; S^4) = -4a_\gamma = -31/45$. Two derivations agree: the integrated trace-anomaly route ($-a_\gamma \int_{S^4} E_4/16\pi^2 = -4a_\gamma$) and the spectral decomposition of the Hodge Laplacian on transverse 1-forms with Faddeev–Popov ghost subtraction [7, 11]. Each unit of Euler characteristic contributes $-2a_\gamma$; $\chi(S^4) = 2$ gives $-4a_\gamma$.
- R2.** $\zeta(0; \text{DtN}; \text{scalar}; S^3) = -1$. The scalar DtN operator on the equatorial S^3 of S^4 (harmonic extension into the cap) has spectrum $\lambda_\ell = \ell(\ell+2)/(\ell+1) = (\ell+1) - 1/(\ell+1)$ ($\ell \geq 1$) with multiplicity $(\ell+1)^2$ (the constant $\ell=0$ mode is the kernel); with $n = \ell+1$ the leading term $\sum_{n \geq 2} n^{2-s}$ continues to $\zeta_R(s-2) - 1$ and the subleading $(1 - 1/n^2)^{-s}$ corrections all carry a factor s and vanish at $s=0$, giving $\zeta(0) = \zeta_R(-2) - 1 = -1$ [11]. (The flat-ball value $\lambda_\ell = \ell$ does *not* apply to the S^4 cap; it would give $-2/3$.)
- R3.** $\zeta(0; \text{DtN}; 1\text{-form}; S^3) = 1$.
- R4.** $\zeta(0; \text{DtN}; \text{Maxwell}; S^3) = 3 = \dim S^3$ (gauge-fixed Maxwell DtN).
- R5.** $\zeta(0; \text{Maxwell}; S^3) = 1$ (the photon-specific equatorial identity).
- R6.** BFK identity verified: with the closed value R1 and the two consistent bookkeepings of Eq. (5) (Section 5.2), $2\zeta(0; D^4) - \zeta(0; \text{DtN}) = -31/45$.
- R7.** Photon Casimir energy on S^3 : $E_{\text{Cas}} = 11/120$.
- R8.** Functional determinant $\det' \Lambda_{S^3}^{\text{Maxwell}} = 1/(2\pi^3)$.
- R9.** $\eta(\Lambda_{S^3}^{\text{transverse 1-form DtN}}) = \zeta(0) = 1$, conformally invariant (the one-form constituent of R8; the full Maxwell DtN has $\zeta(0) = 3$). (Distinct from the de Rham APS eta $\eta_{\text{APS}}(S^3) = 0$, which enters the topological quantization of Section 11; they are different operators on the same S^3 .)

- R10.** A Robin condition $(\partial_n + \alpha)u = 0$ at the deformed equator shifts the DtN spectrum by $\alpha = \tan a_\gamma \approx a_\gamma$ — the analytic origin of the cap-saddle drive.
- R11.** Under the Robin shift, $\zeta(0; \Lambda_{\text{Robin}})$ shifts linearly in α .
- R12.** $\int_{D^4} Q_4 / \int_{D^4} E_4 = 1/4$ exactly, fixing the Q -vs- E normalization.

5.2 The two consistent BFK bookkeepings

The hemisphere zeta value carries a nontrivial spectral weight, and the BFK identity (5) admits two internally consistent bookkeepings that differ only in how the gauge/ghost modes are assigned between the hemisphere and the DtN operator. Both return the closed value $\zeta(0; S^4) = -31/45$.

$$(a) \text{ scalar-DtN split: } \zeta(0; D^4) = -2a_\gamma - \frac{1}{2} = -\frac{38}{45}, \quad 2\left(-\frac{38}{45}\right) - (-1) = -\frac{31}{45} \quad \checkmark \quad (6)$$

$$(b) \text{ gauge-fixed Maxwell split: } \zeta(0; D^4) = \frac{52}{45}, \quad 2\left(\frac{52}{45}\right) - 3 = -\frac{31}{45} \quad \checkmark \quad (7)$$

The equivalence of the two bookkeepings is itself a check on the spectral arithmetic; the gauge-fixed Maxwell DtN value 3 (R4) and the symmetric hemisphere value $52/45$ are used in the boundary analysis of Section 12. These spectral results are *proven*; the closed value $\zeta(0; \text{Maxwell}; S^4) = -31/45$ is itself *derived* from the Gilkey heat-kernel coefficient in Section 3 (cross-checked against Christensen–Duff 1980; Esposito 1995, in the conventions of [11]). The photon-specific value $\zeta(0; \text{Maxwell}; S^3) = 1$ (R5), together with the DtN data, underlies the cap-variational restoring force of Section 12.

6 Spectral calculations: R1–R12

This section collects the explicit computations behind the twelve spectral results R1–R12 stated above.

R1: $\zeta(0; \text{Maxwell}; S^4) = -31/45$. In Lorenz gauge with Faddeev–Popov ghosts the Maxwell Laplacian on transverse one-forms of the unit S^4 has spectrum $\lambda_\ell = \ell(\ell + 3) + 2$ ($\ell \geq 1$) with degeneracy $d_\ell = 2\ell(\ell + 3)(2\ell + 3)/3$; the complex ghost contributes a scalar Laplacian with spectrum $\ell(\ell + 3)$. The combined zeta function evaluates at $s = 0$ to $\zeta(0; \text{Maxwell}; S^4) = -31/45 = -4a_\gamma$, reflecting $\int_{S^4} E_4 = 64\pi^2$ and $\int_{S^4} \langle T^\mu{}_\mu \rangle = 4a_\gamma$.

R5: $\zeta(0; \text{Maxwell}; S^3) = 1$. On S^3 the transverse Maxwell spectrum is $\mu_\ell = (\ell + 1)^2$ with degeneracy $2\ell(\ell + 2) = 2(\ell + 1)^2 - 2$. With $n = \ell + 1$, $\zeta(s; S^3) = 2\zeta_H(2s - 2, 2) - 2\zeta_H(2s, 2)$, and the Hurwitz values $\zeta_H(-2, 2) = -1$, $\zeta_H(0, 2) = -\frac{3}{2}$ give $\zeta(0; \text{Maxwell}; S^3) = 2(-1) - 2(-\frac{3}{2}) = 1$. Scalars and Dirac spinors give 0; the value 1 is the photon-specific signature on S^3 that supplies the cap restoring force.

R8: DtN determinant $\det' \Lambda_{S^3}^{\text{Maxwell}} = 1/(2\pi^3)$. The gauge-fixed Maxwell DtN is the combination $\Lambda^{\text{Maxwell}} = \Lambda^{\text{1-form}} - 2\Lambda^{\text{scalar ghost}}$ (transverse one-form minus two Faddeev–Popov scalar ghosts), *not* a single one-form spectrum. Its two constituents are: the transverse one-form DtN with $\mu_\ell = \ell + 1$, degeneracy $2\ell(\ell + 2)$ ($\zeta(0) = 1$, $\zeta'(0) = 2\zeta'_H(-2, 2) + \log 2\pi$); and the scalar-ghost DtN with $\mu_\ell = \ell(\ell + 2)/(\ell + 1)$, multiplicity $(\ell + 1)^2$ ($\zeta(0) = -1$, $\zeta'(0) = \zeta'_H(-2, 2) - \log \pi$, the same operator as R2). Hence $\zeta(0; \text{Maxwell}) = 1 - 2(-1) = 3$ (R4) and $\zeta'_\Lambda(0) = [2\zeta'_H(-2, 2) + \log 2\pi] - 2[\zeta'_H(-2, 2) - \log \pi] = \log(2\pi^3)$ (the Hurwitz term $\zeta'_H(-2, 2)$ cancels exactly), i.e. $\det' \Lambda_{S^3}^{\text{Maxwell}} = 1/(2\pi^3) \approx 0.01613$, confirmed numerically to 80+ digits. (Pairing the one-form spectrum $\mu_\ell = \ell + 1$ alone with this determinant is incorrect: the one-form alone gives $\det' \approx 0.169$, not $1/(2\pi^3)$.)

R9: η -invariant $\eta(\Lambda_{S^3}^{\text{transverse 1-form}}) = 1$. The transverse one-form DtN eigenvalues $\mu_\ell = \ell + 1 > 0$ are all positive, so its eta-invariant equals its zeta value $\zeta(0; \Lambda^{\text{1-form}}) = 1$; it is conformally invariant. (This is the one-form constituent of R8, $\zeta(0) = 1$; the full gauge-fixed Maxwell DtN has $\zeta(0) = 3$, R4.)

R12: $\int_{D^4} Q_4 / \int_{D^4} E_4 = 1/4$. By Chang–Yang $\int_{D^4} Q_4 = 8\pi^2$ and by Gauss–Bonnet–Chern with boundary corrections $\int_{D^4} E_4 = 32\pi^2$, giving the ratio $1/4$. The structural identity $Q_4 = E_4/4 +$ (conformal divergence) on conformally flat manifolds underlies the resonance $4Q_4 = \mu_1(P_4) = 24$ on S^4 .

7 BFK decomposition and the DtN calculation

The Burghlelea–Friedlander–Kappeler formula reads $\log \det' \Delta_M = \log \det' \Delta_{M_+}^{\text{Dir}} + \log \det' \Delta_{M_-}^{\text{Dir}} - \log \det' \Lambda_\Sigma$. For $S^4 = D_+^4 \cup_{S^3} D_-^4$ with $D_+^4 \cong D_-^4$, separation of variables in S^3 harmonics gives, for a transverse one-form harmonic with $\Delta_{S^3} \omega_\ell = (\ell+1)^2 \omega_\ell$, the radial problem $[-\partial_\theta^2 - 2 \cot \theta \partial_\theta + (\ell+1)^2 / \sin^2 \theta] u_\ell = \lambda u_\ell$. The harmonic extension $u_\ell \propto (\sin \theta)^{\ell+1}$ yields the transverse one-form DtN spectrum $\Lambda_{S^3} \omega_\ell = (\ell+1) \omega_\ell$ ($\zeta(0) = 1$). The gauge-fixed Maxwell determinant is then the combination $\Lambda^{\text{1-form}} - 2\Lambda^{\text{scalar ghost}}$, giving $\zeta'_\Lambda(0) = \log(2\pi^3)$ and $\det' \Lambda_{S^3}^{\text{Maxwell}} = 1/(2\pi^3)$ (R8); the one-form spectrum alone does *not* give this determinant. This is consistent with the $\zeta(0)$ bookkeeping: the scalar-ghost split uses the scalar DtN $\zeta(0) = -1$ and the hemisphere value to give $2\zeta(0; D^4) - \zeta(0; S^4) = 2(-38/45) - (-31/45) = -1$ (bookkeeping (a) of Section 5), while the gauge-fixed Maxwell DtN itself has $\zeta(0) = 3$ (R4, bookkeeping (b)) — two distinct operators that must not be conflated.

8 The magnitude problem: Kaloper–Padilla sequestering

The first face of the cosmological constant problem — why the gravitating vacuum energy is 120 orders of magnitude below the naive estimate — is addressed by vacuum-energy sequestering (Axiom A2, [12, 14]). In the Kaloper–Padilla construction the classical action is supplemented by a global constraint that promotes the cosmological counterterm to a dynamical variable. Integrating out that variable enforces

$$\Lambda_{\text{eff}} = \frac{1}{4} \frac{\int \sqrt{g} \langle T^\mu{}_\mu \rangle d^4x}{\int \sqrt{g} d^4x}, \quad (8)$$

i.e. the effective cosmological constant is the spacetime average of the matter stress-tensor trace. The radiative quartic vacuum energy, being a spacetime constant, drops out of the *difference* between the local source and its global average: it does not gravitate. This is the standard sequestering resolution of the magnitude problem.

Sequestering is internal to CSG, not an external add-on. The constraint (8) need not be imposed by hand: it is generated by the conformal structure of CSG itself. In Weyl geometry the gravitational action $S = \int \sqrt{g} (\alpha \tilde{C}^2 + \beta \tilde{R}^2)$ decomposes, on fixing the Weyl gauge ($\tilde{R} = R + 6\nabla_\mu \omega^\mu + 6\omega_\mu \omega^\mu$), into a Weyl-squared term plus a scalar: defining $\varphi \equiv \sqrt{2\beta} (R + 6\nabla \cdot \omega + 6\omega^2)$ turns the R^2 term into $\frac{1}{2}\varphi^2$, and integrating out the Weyl gauge field ω_μ supplies its kinetic term, giving $S = \int \sqrt{g} [\alpha C^2 + \frac{1}{2}(\partial\varphi)^2 - V(\varphi) + \mathcal{L}_m]$. The compensator φ is the Stückelberg field of the broken Weyl symmetry, and it plays exactly the role of the Kaloper–Padilla dilaton: the Stückelberg identity by which ω_μ absorbs the scalaron is, term by term, the KP global constraint, both being realizations of the same broken-CFT structure. Spontaneous breaking by the scalaron VEV generates the Planck mass, $M_{\text{Pl}}^2 = v_\varphi^2/8\pi$, and fixes the residual $\Lambda_{\text{eff}} = V(v_\varphi)$; every radiative correction $\delta\Lambda$ is quadratic in δv_φ and reabsorbed by the scalaron at the same

order in $\hbar$ (the linear term is forbidden by the minimum condition), so Λ_{eff} is radiatively stable. Sequestering is therefore *derived within CSG* (the Built-in Sequestering proposition — a proof sketch by direct identification, with full all-orders rigor of the scalaron–dilaton map and of the Stückelberg–KP correspondence in non-trivial backgrounds the remaining refinement), not an adopted external mechanism; and because CSG carries no propagating graviton loop, it evades the standard graviton-loop objection to KP. Crucially for what follows, Eq. (8) is also the seed of the value problem: it expresses Λ_{eff} in terms of a global average that itself depends on Λ_{eff} through the geometry, so the equation is implicit and must be solved self-consistently. That self-consistency is the subject of the next section.

On the Euclidean hemisphere the same averaging, applied to the photon trace anomaly, yields the numerator of the topological quotient: with $\int_{D^4} = \frac{1}{2} \int_{S^4}$ and $\int_{S^4} \langle T^\mu{}_\mu \rangle = 4a_\gamma$ (Section 2), $\frac{1}{2} \Lambda_{\text{eff}} V_4(D^4) = a_\gamma$, which combines with the Q -charge denominator $C_Q = 8\pi^2$ in Section 11.

9 Causal legitimation of the finite region and the saddle scale

Two inputs above were stated as model-building choices: the restriction of the averaging to the finite hemisphere D^4 rather than a global region requiring an observer measure, and the identification of the saddle scale with $L_{\text{HH}} = 1/H_0$. Both follow from a causal variational principle. Shaw and Barrow [13] promote Λ to a field and impose stationarity of the classical action under variations confined to the causal past of a measurement — the past light cone M :

$$\frac{dI_{\text{class}}(\Lambda; M)}{d\Lambda} = 0. \quad (9)$$

The region M needs no anthropic weighting: it is what a local measurement can access. The past light cone of an observer in an open FLRW foliation is topologically a four-ball, so M is the Lorentzian counterpart of the Euclidean hemisphere D^4 on which the BFK decomposition operates. The hemisphere is therefore not an arbitrary cutoff but the causal region itself, and the boundary that BFK requires is the boundary of that region.

The saddle scale, parameter-free. With extrinsic curvature $\text{tr} K \sim H$ on the bounding surface and four-volume $V_M \sim t_U A_{\partial M}$, Eq. (9) fixes the order of magnitude

$$\Lambda \sim \frac{H_0 A_{\partial M}}{V_M} \sim t_U^{-2}, \quad (10)$$

so $L_{\text{HH}} \sim 1/H_0$ with no free scale. Solving the self-consistency $\Lambda = t_U^{-2}$ numerically gives $\Omega_\Lambda \approx 0.49$ at factor-unity normalization, and at the observed $\Omega_\Lambda = 0.685$ the time ratio $t_\Lambda/t_U = 0.73$ — the geometric origin of the coincidence. The mechanism fixes the magnitude and the coincidence to order unity; it does *not* fix the precise value of Ω_Λ (Section 25).

Structural identity with the cap mechanism. Equation (9) and the cap-variational principle of Section 12 are one variational statement in two realizations: a bulk trace drive balanced against a boundary restoring force over a finite causal region. Shaw–Barrow is the Lorentzian, dynamical version, driven by the classical baryon trace $\zeta_b = 1 - M_q/M_N \approx \frac{1}{2}$; the cap mechanism is the Euclidean, topological version, driven by the photon anomaly $3a_\gamma$ against the DtN restoring force. The two differ decisively in which trace enters: Shaw and Barrow set the photon contribution to the matter Lagrangian to zero classically ($E^2 = B^2$ for radiation), which is exactly where the conformal anomaly — the quantum correction they omit — resides. By the Komargodski–Schwimmer infrared selection (Section 2) the baryon trace decouples in the deep infrared, so the anomaly is the surviving drive.

Why the realization must be topological. A Lorentzian light-cone evaluation of the anomaly fails quantitatively: the anomaly stress $\sim a^4 H^4$ is weighted to early times in the light-cone integral, and the dynamical (Riegert/DtN-determinant) attempt to derive $a_\gamma/(8\pi^2)$ is too weak by the four-volume factor $(M_{\text{Pl}}/H)^4 \sim 10^{240}$ (Section 14, Mechanism 2). Only the topological evaluation — the a_4 heat-kernel drive against the scale-invariant DtN $\zeta(0) = 3$, normalized by the Chang–Yang charge $C_Q = 8\pi^2$ — yields the parameter-free ratio. The causal principle legitimizes the finite region and the scale; the topological character of the quotient is what makes the anomaly evaluation finite and unique.

10 The value problem: a self-consistency theorem

Sequestering removes the radiative vacuum energy but leaves the residual scale to be fixed. We now show that the residual Λ is not free: it is the unique fixed point of the averaging map (8).

10.1 The self-consistency iteration

Proposition 2 (Kaloper–Padilla self-consistency). *Let the effective cosmological constant be updated by the four-volume average of the total energy density over a closed cosmological history, $\Lambda^{(n+1)} = \frac{1}{4}\langle\rho_{\text{total}}\rangle_{V_4}^{(n)}$. Then on a closed cosmology ($k = +1$) with finite four-volume $V_4 = \int dt a^3 < \infty$, the update map is a contraction with a unique fixed point Λ^* , and Λ^* is the unique value satisfying the trace-zero condition.*

Proof. Two complementary parts. *Convergence (analytic contraction).* Parametrize the update map by Λ via the Friedmann constraint at $a = 1$, so that one iteration is $\mathcal{F}(\Lambda) = \frac{1}{4}\langle\rho_{\text{total}}(\Lambda)\rangle_{\text{ren}}$ with the four-volume (a^3 -weighted) average. Writing the trace function $G(\Lambda) \equiv \frac{1}{4}\langle\rho_{\text{total}}\rangle - \Lambda = -\frac{\kappa}{4}\langle T \rangle$, the derivative is $\mathcal{F}'(\Lambda) = 1 + G'(\Lambda)$, so the Lipschitz constant is $L = |1 + G'(\Lambda)|$. Increasing Λ compresses the closed history (shorter recollapse time, smaller a_{max}), which strictly reduces the a^3 -weighted average of the positive matter and kinetic densities; hence G is strictly decreasing and $G'(\Lambda) < 0$. The contraction condition $L = |1 + G'(\Lambda)| < 1$ then holds *provided* $G'(\Lambda) > -2$. The sign $G' < 0$ is established analytically (the compression argument above); the bound $|G'| < 2$ is established *numerically* ($|G'| \approx 0.7$, hence $L \approx 0.13$ – 0.18 in the full pipeline, $L \approx 0.3$ in the reduced model). By the Banach fixed-point theorem the iteration then has a unique attractor. A fully analytic bound on $|G'|$ remains the open refinement, so the contraction is *numerically established* rather than closed by an analytic proof. *Correctness:* the self-consistency condition $\Lambda^* = \frac{1}{4}\langle R \rangle_{\text{ren}}$ is, by the trace identity $R = 4\Lambda + \kappa T$, algebraically equivalent to

$$\langle T(\Lambda^*) \rangle_{\text{ren}} = 0, \quad (11)$$

precisely the Kaloper–Padilla sequestering condition. Existence requires $k = +1$ with $V_4 < \infty$; in flat or open cosmologies V_4 diverges and the volume average is ill-defined. Since $\langle -T \rangle(\Lambda)$ is continuous, positive for small Λ and negative for large Λ (the same monotonicity), the fixed point is unique by the intermediate value theorem. The independent computation of Section 21 (M1- Λ) confirms the trace-zero identity symbolically, the finiteness of V_4 for $k = +1$ against divergence for $k = 0, -1$, and the existence of a unique negative fixed point. $\square$

10.2 Compatibility of a closed history with open observation

Proposition 2 requires a closed Euclidean structure with finite four-volume, which at first sight conflicts with the open ($\Omega_K > 0$) Lorentzian patch predicted in Section 11. There is no conflict, because three distinct regimes are involved: (i) the *full cosmological history* (Big Bang $\rightarrow$ expansion $\rightarrow$ recollapse $\rightarrow$ Big Crunch), closed ($k = +1$), supplying the finite V_4 ; (ii) the *present observable patch*, the Wick-rotated cap-shifted instanton, hyperbolic ($K < 0$, $\Omega_K > 0$) at the present epoch; (iii) the *Euclidean instanton* $D^4 \subset S^4$, compact, on which the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/(8\pi^2)$ is

fixed. These live on different manifolds related by analytic continuation and fix complementary quantities; their coexistence is the content of the no-boundary construction (Axiom A1).

10.3 The Coleman–Weinberg vacuum and the status of the absolute value

The fixed point can equivalently be read off the Coleman–Weinberg effective potential of the scalaron, $V(\varphi) = A\varphi^4(\ln(\varphi^2/v_S^2) - \frac{25}{6})$ with $A \propto a_\gamma$. Its unique nontrivial stationary point sits at $\varphi_{\min} = v_S e^{11/6}$ (verified in Section 21, M1-CW), and in the stable convention ($V_{\min} < 0$) this is a minimum with negative Λ^* . What Proposition 2 establishes *structurally* is therefore that Λ is not a free parameter: its sign is negative, the closed-history requirement and the trace-zero characterization are proven, and the iteration converges.

What it does *not* do unconditionally is pin the *sharp* absolute value, and the reason is instructive. The average is taken over ρ_{total} , not ρ_m : the scalaron potential component does not dilute, so $\langle \rho_{\text{total}} \rangle_{V_4}$ is *a_{max}-stable* (it tends to the non-diluting plateau as $a_{\text{max}} \rightarrow \infty$), whereas averaging ρ_m alone would spuriously collapse as $1/a_{\text{max}}^3$ — a category error to be avoided. This *a_{max}*-stability of the total average and the topological constraint $\Lambda_{\text{eff}} V_4 = 4a_\gamma$ are solid.

Correction: the volume route does not deliver the sharp value. An earlier sharpness analysis wrote $\Lambda^* = \frac{1}{4}\langle R \rangle_{V_4}$ for a closed matter+ Λ history and reported that a *moderate* a_{max} breaks the de Sitter identity (giving $\langle R \rangle/4 \approx 0.76\text{--}0.89 \times 3\Omega_\Lambda$) while a long plateau returns the identity $\Lambda_{\text{eff}} = \Lambda$. That construction is *geometrically inconsistent* and its numbers are artefacts: it fixes a constant positive Ω_Λ and then adds a curvature term forcing $H^2 = 0$ at a_{max} , but a universe with constant $\Omega_\Lambda > 0$ never recollapses, so H^2 becomes *negative* on $1 < a < a_{\text{max}}$ and was silently floored. The correct treatment requires the genuine recollapse driver $V(\varphi) < 0$. Integrating the full Friedmann+Klein–Gordon system with the Coleman–Weinberg potential and feeding $-\Lambda^*/3$ back into the fixed point yields a sharper, and more sobering, picture:

- Λ^* is *universal*: independent of φ_0 and of Ω_m (matter dilutes out of the V_4 -weighted average). This universality is a genuine structural feature.
- Its *value* is amplitude-set: $\Lambda^* \approx -0.10$ in $3H_0^2$ units for the fixed-point amplitude $A_{\text{CW}} = a_\gamma/128\pi^2$ (and ≈ -0.002 for the M5 amplitude), *not* the -0.691 quoted from the proxy.
- $\Omega_\Lambda = (V(\varphi_0) - \Lambda^*)/3$ is *not* fixed: it varies fully with φ_0 . KP self-consistency pins Λ^* but leaves φ_0 — the scalaron value “today” — free, so the absolute Ω_Λ is not determined by this route.

The honest status is therefore: Λ is not a free parameter and its sign is fixed (structural, proven); the bulk vacuum cancels by sequestering, removing the 10^{122} hierarchy; but the volume/self-consistency route delivers the universality *structure* and the order of magnitude only, neither the sharp Λ^* nor the absolute Ω_Λ . Sharpening to $\Omega_\Lambda \approx 0.704$ requires the halo-weighted prior or the causal closure of Section 16.2. None of this touches the theorem-level, prior-free output — the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ — which is independent of the absolute scale.

10.4 The magnitude of the curvature: coherent spectral accumulation

The previous subsection settled what the volume/self-consistency route does and does not fix. One question it deliberately left in words is the *magnitude* question for the testable output: after sequestering removes the bare M_{Pl}^4 vacuum, the renormalized photon trace anomaly is still only a *local* de Sitter density

$$\rho_{\text{anom}} = \frac{1}{4}\langle T^\mu{}_\mu \rangle = \frac{a_\gamma}{8\pi^2} 3H^4, \quad \frac{\rho_{\text{anom}}}{\rho_{\text{crit}}} = \frac{a_\gamma}{8\pi^2} \frac{H^2}{M_{\text{Pl}}^2} \sim 10^{-122}, \quad (12)$$

so a naive reading would predict $\Omega_K \sim 10^{-125}$, not $\sim 10^{-3}$. A factor $(M_{\text{Pl}}/H)^2$ must lift the local density to the curvature scale. We record here the mechanism that supplies it and state precisely how far it reaches.

Coherence is the linearity of the spectral sum. The gauge-fixed Maxwell operator on dS_4 has the discrete spectrum (Allen 1986; Higuchi 1991) $\lambda_n = n(n+2)H^2$ with multiplicity $d_n = 2n(n+2)$, $n \geq 1$. The vacuum sum runs over these *fixed* eigenmodes, $\sum_n d_n$, and is therefore *linear* and deterministic: this linearity *is* the $\times N$ coherence. The familiar $\times \sqrt{N}$ estimate applies only to *stochastic*, phase-uncorrelated contributions; the spectral modes on the compact sphere are phase-locked by the geometry, so the ζ -regularized sum over them is inherently coherent. The coherence is thus not an added postulate but a property of the spectrum. The bulk mode count to the cutoff $n_{\text{max}} = M_{\text{Pl}}/H$ is

$$N_{\text{tot}}(n_{\text{max}}) = \sum_{n=1}^{n_{\text{max}}} 2n(n+2) = \frac{2}{3}n_{\text{max}}^3 + 3n_{\text{max}}^2 + \frac{7}{3}n_{\text{max}} \sim \frac{2}{3}\left(\frac{M_{\text{Pl}}}{H}\right)^3, \quad (13)$$

and the horizon (area) projection reduces the volume scaling to $N_{\text{eff}} \sim (M_{\text{Pl}}/H)^2 = S_{dS}/\pi$, the de Sitter entropy. The $(M_{\text{Pl}}/H)^2$ *scaling* is rigorous; its $O(1)$ prefactor is convention-dependent (mode basis, projection scheme).

What the coherent lift delivers. Multiplying the local density (12) by the coherent mode number gives, exactly,

$$\frac{N_{\text{eff}} \rho_{\text{anom}}}{\rho_{\text{crit}}} = \frac{a_\gamma}{8\pi^2}, \quad (14)$$

i.e. precisely the dimensionless ratio Ω_K/Ω_Λ . The 10^{122} gap is closed *structurally* for the quantity that the theory actually predicts and that DESI actually measures: the coherent spectral sum lifts the local anomaly to the $O(10^{-3})$ curvature scale. This is the magnitude resolution for the testable prediction, and it is independent of the absolute-value question.

From the ratio to the absolute value. Promoting the ratio (14) to the absolute $\Omega_\Lambda = 4a_\gamma$ requires one further topological factor, and that factor is not free:

$$\frac{\rho_\Lambda}{\rho_{\text{anom}}} = 32\pi^2 \left(\frac{M_{\text{Pl}}}{H}\right)^2 = \left(\int_{D^4} E_4\right) N_{\text{eff}}, \quad (15)$$

with $\int_{D^4} E_4 = 32\pi^2$ the hemisphere Euler number already used in the budget ladder. Equivalently $\Omega_\Lambda = |\zeta(0; \text{Maxwell}; S^4)| = 4a_\gamma$, the cutoff-free integrated trace anomaly.

A second, independent anchor of the value: the conformal scale-flow. The number $4a_\gamma$ is not only the spectral determinant $|\zeta(0)|$; it is also the conformal scale-anomaly coefficient of the free energy. Under a global Weyl rescaling $g \rightarrow e^{2\sigma}g$ the anomaly-induced (Riegert–Mottola) action obeys

$$\frac{dS_{\text{anom}}}{d\sigma} = \frac{a_\gamma}{16\pi^2} \int_{S^4} \sqrt{g} E_4 = 4a_\gamma, \quad (16)$$

because $\int_{S^4} E_4 = 64\pi^2$ is a topological invariant. Thus $4a_\gamma$ is fixed *cutoff-free by two independent routes* — the spectral determinant and the scale-flow — neither of which uses the N_{eff} bulk projection. Consequently the $O(1)$ prefactor of that projection (the heuristic ambiguity $4/3\pi$ vs 1) is *irrelevant to the value*: it is bypassed, not tuned. What remains hypothesis-level is only the *identification* of Ω_Λ with this invariant. The scale-flow physically motivates it, but does not force it dynamically: $S_{\text{anom}}(\sigma) = 4a_\gamma \sigma$ is linear, so it has no σ -saddle, and the dynamical fixing of σ (the M5 route) returns $1/12a_\gamma \approx 0.48$, not 0.69. The identification is therefore the same conjecture flagged for A5 and in the budget ladder, now with a physical anchor for its magnitude.

Honest status of this subsection. *Solved (structural):* the coherence (as the linearity of the fixed-mode spectral sum), the $(M_{\text{Pl}}/H)^2$ scaling, and the magnitude of the curvature ratio (14) — the 10^{122} gap is closed for the testable Ω_K prediction; the bridge factor $32\pi^2 = \int_{D^4} E_4$ is geometric, not a knob; and the *value* $4a_\gamma$ is fixed cutoff-free by the two independent anchors above. *Open (hypothesis-level, and not decidable by DESI):* (a) the *identification* $\Omega_\Lambda = 4a_\gamma$ — physically anchored by the scale-flow but not forced by a saddle (the dynamical route gives 0.48); and (b) the epoch binding, since $\Omega_\Lambda(z)$ is epoch-dependent, $4a_\gamma$ is matched only at $z \approx 0$, and no dynamical attractor sits at $4a_\gamma$ — the coincidence problem (iii), here inherited and addressed only by observer-selection ($\Omega_\Lambda \approx 0.704$, Section 16.2), not by a timeless mechanism. The ratio $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ remains untouched by both, as it is independent of the absolute scale.

11 The parameter-free relation: the eight-step derivation

We now assemble the topological quotient linking the residual cosmological constant to the spatial curvature. The first six steps and the two Euclidean sub-steps of the seventh are theorem-level; the one remaining genuinely chosen sub-step is the data-anchored reference state A5c.

Step 1 (anomaly integral). By Eq. (2) and Gauss–Bonnet–Chern, $\int_{S^4} \langle T^\mu{}_\mu \rangle = \frac{a_\gamma}{16\pi^2} \int_{S^4} E_4 = 4a_\gamma$.

Step 2 (sequestering on the hemisphere). By Eq. (8) and $\int_{D^4} = \frac{1}{2} \int_{S^4}$, $\frac{1}{2} \Lambda_{\text{eff}} V_4(D^4) = a_\gamma$.

Step 3 (Q -charge). By Lemma 1, $C_Q = 8\pi^2$.

Step 4 (Wess–Zumino consistency). Under $g \rightarrow e^{2\sigma} g$ the anomaly action obeys the Wess–Zumino consistency relation [15], with linear response coefficient $a_\gamma C_Q/(16\pi^2) = a_\gamma/2$.

Step 5 (hemisphere quotient). The dimensionless quotient

$$\mathcal{R} \equiv \frac{a_\gamma}{C_Q} = \frac{a_\gamma}{8\pi^2} = \frac{31}{1440\pi^2} \quad (17)$$

contains no scale, no Newton constant, no Planck mass.

Step 6 (photon selection). By Axiom A3 and the 26-mechanism exhaustion (Section 14), the coefficient is the photon’s $a_\gamma = 31/180$. This step uses the BFK consistency check of Section 5 but, as emphasized in Section 12, the derivation is otherwise *boundary-free*.

Step 7 (Janus-point matching). Three logically distinct sub-components:

- (7a) *Cap-saddle construction* (Section 12): a variational principle on D^4 produces a saddle at $\delta\theta^* = a_\gamma$ to leading order. Euclidean, geometric. *Proven*.
- (7b) *Wick rotation* (Bucher–Goldhaber–Turok open-universe template [16, 17]): translates the Euclidean cap-saddle into a Lorentzian Friedmann patch with curvature $\Omega_K^{\text{conformal}}$. *Proven*.
- (7c) *Identification A5*: relates $\Omega_K^{\text{conformal}}$ to the Friedmann observable Ω_K^{obs} . Its Euclidean half A5a is a *theorem* and the Wick-rotation kinematics A5b are *derived* given B1; the residual is the data-anchored reference choice A5c. *Specification (the only genuinely chosen element)*.

Under (7a)+(7b)+(7c) the dimensionless Euclidean ratio projects onto the present-epoch observable, giving Eq. (1).

Step 8 (Friedmann translation). With $\Omega_m + \Omega_\Lambda + \Omega_K = 1$ and $\Omega_\Lambda = \mathcal{L} |\Omega_K|$, $\mathcal{L} = 8\pi^2/a_\gamma = 458.4$,

$$|\Omega_K| = \frac{1 - \Omega_m}{\mathcal{L} \pm 1}, \quad \Omega_\Lambda = \mathcal{L} |\Omega_K|, \quad (18)$$

the $+$ sign holding for an open universe. With the ladder-consistent budget $\Omega_m = 0.3096$ (Section 16.1), $|\Omega_K| \approx 1.50 \times 10^{-3}$ and $\Omega_\Lambda = 4a_\gamma = 0.6889$, matching the robust Planck 2018 (TT,TE,EE+lowE+lensing+BAO) value $\Omega_\Lambda = 1 - \Omega_m = 0.6889$ ($\Omega_m = 0.3111 \pm 0.0056$) [22] exactly to four decimals; the CSG trio $(\Omega_m, \Omega_\Lambda, \Omega_K) = (0.3096, 0.6889, +0.0015)$ is consistent with Planck+BAO including its curvature constraint $\Omega_K = 0.001 \pm 0.002$.

Postulate 1 (A5: Anomaly–Curvature Correspondence). The conformal-anomaly-derived curvature $\Omega_K^{\text{conformal}}$ obtained from the hemisphere quotient (17) via the cap-saddle and Wick rotation equals the observed Friedmann spatial-curvature parameter Ω_K^{obs} at the present epoch.

A5 is not monolithic. Following the decomposition adopted here, its Euclidean half — A5a, the topological identity $\Omega_K^{\text{conformal}} = a_\gamma/C_Q$ — is a *theorem* (Cauchy functional equation for linearity, Chern–Weil basis change for the $1/8\pi^2$, and the APS index theorem on the de Rham complex), up to the bounded BFK boundary refinement $|\epsilon_\partial| \leq 1.5\%$. What remains a physical postulate is the identification A5b/A5c: that this Euclidean quantity, carried through the Wick rotation (A5b, derivable in saddle-point approximation given B1), equals the present-epoch Friedmann observable under the canonical Λ -dominated reference (A5c). Thus everything in Eq. (1) except the A5b/A5c identification is theorem-level or literature-supported.

12 The cap-variational mechanism, stability, and the boundary refinement

12.1 The saddle equation

The hemisphere “breathes” by an angular shift $\delta\theta$ of the polar cap. The photon anomaly supplies a bulk drive linear in $\delta\theta$ at the equator (Robin shift $\alpha = \tan a_\gamma \approx a_\gamma$, Spectral Result R10), and the DtN determinant supplies a restoring force. Balancing them gives the leading saddle $\delta\theta_{\text{lin}}^* = a_\gamma$, and the exact angular dependence gives

$$a_\gamma \cos^4 \delta\theta = \sin \delta\theta, \quad \implies \quad \delta\theta_{\text{exact}}^* = 0.16391, \quad (19)$$

a $\sim 5\%$ downward shift (cubic truncation $\delta\theta_{(3)}^* = a_\gamma - 2a_\gamma^3$).

12.2 Stability

The second variation around the cap-saddle is positive in every inhomogeneous sector, and this is established *analytically*, not merely by mode-by-mode shooting. The deformed cap $D^4(\theta_0 = \pi/2 + \delta\theta^*)$ with $\delta\theta^* = 0.164$ is a geodesic ball on S^4 with $\theta_0 < \pi$, hence a proper subdomain. The fluctuation operator is the Hodge–de Rham Laplacian (with the fixed S^4 curvature potential, the same on cap and hemisphere), so its Dirichlet eigenvalues obey *domain monotonicity*: $\lambda_{\text{Dir}}(\theta_0)$ is strictly decreasing in θ_0 , with limit the closed- S^4 eigenvalue as $\theta_0 \rightarrow \pi$. The closed- S^4 inhomogeneous spectra are positive in closed form — scalars $\ell(\ell+3) \geq 4$, coexact 1-forms and transverse-traceless tensors $(\ell+1)(\ell+2) \geq 6$ and ≥ 12 respectively — so for every inhomogeneous mode

$$\lambda_{\text{Dir}}\left(\frac{\pi}{2} + \delta\theta^*\right) > \lambda_{\text{closed}}(S^4) > 0.$$

The shooting computation confirms the bracketing exactly (e.g. scalar $\ell = 1$: hemisphere 10.0 $\rightarrow$ cap 8.16 $\rightarrow$ closed limit 4; TT $\ell = 2$: 20.0 $\rightarrow$ 17.27 $\rightarrow$ 12), but the positivity itself follows from the two theorems (domain monotonicity and the positive closed spectrum), not from the numerics.

The transverse 1-form is distinguished from the scalar by the Hodge–Weitzenböck term $\Delta_{\text{Hodge}}^{(1)} = \Delta_{\text{Bochner}} + 3$ on S^4 . The only non-positive direction is the constant ($\ell = 0$) conformal/lapse mode, of Morse index 1, handled by the Gibbons–Hawking–Perry contour rotation (Section 18); the saddle is thus genuine. Reconfirmed independently in Section 21 (M1-H).

12.3 The boundary refinement and why it cannot move the main result

The cap-shifted boundary carries Branson–Gilkey heat-kernel coefficients. Two facts make their effect controlled. First, a *rigorous* BFK symmetry: writing $\zeta_{\text{bnd}}(\pi/2 + \delta) - 3/2 \equiv g(\delta)$, the gluing identity forces $\zeta_{\text{bnd}}(\pi/2 + \delta) + \zeta_{\text{bnd}}(\pi/2 - \delta) = 3$ for all δ , so g is an *odd* function and all even-order Taylor coefficients vanish identically. Second, and decisively, the main derivation (Steps 1–8) uses only the closed- S^4 anomaly integral a_γ and the Q -charge $C_Q = 8\pi^2$; neither couples to boundary heat-kernel coefficients. The boundary appears *only* in the BFK consistency check (Step 6) and in the cap-saddle shift. Consequently the worst-case boundary uncertainty is geometrically excluded from the prediction chain $a_\gamma \rightarrow \Omega_K$. Quantitatively, in the natural renormalization scheme ($\mu = R^{-1}$ fixed) the shift is 0% (BFK-protected); in the cap-scale scheme it is bounded at $|\Delta\Omega_K|/\Omega_K \leq 1.5\%$; the unprotected worst case ($\leq 49\%$) does not propagate, being excluded by the boundary-free structure of the main derivation.

12.4 The prediction hierarchy L0–L3

Level	Formula	Value	Status
L0	$R_0 = a_\gamma/(8\pi^2)$	2.181×10^{-3}	topological-anomalic (exact theorem)
L1	$R_{\text{cap}} = \delta\theta_{\text{exact}}^*/(8\pi^2)$	2.076×10^{-3}	cap-variational refined
L2	$R_{\text{cap}+\partial} = R_{\text{cap}}(1 + \epsilon_\partial)$	$2.05\text{--}2.11 \times 10^{-3}$	boundary check ($ \epsilon_\partial \leq 1.5\%$)
L3	$\Omega_K^{\text{obs}} = R \cdot \Omega_\Lambda$	$1.42\text{--}1.49 \times 10^{-3}$	present-epoch observable

Table 1: The four-level prediction hierarchy. L0 is the exact topological identity on D^4 . L1 incorporates the exact cap-saddle solution of Eq. (19). L2 is the boundary consistency band (Section 12.3); it does *not* enter the boundary-free L0/L1 derivation. L3 is the Friedmann translation.

Headline statements are quoted at L0; numerical data comparison uses L1 or L3. The L0-vs-L1 split, $\Delta R = 1.05 \times 10^{-4}$ ($\Delta\Omega_K = 0.72 \times 10^{-4}$), is a small refinement whose observational discrimination is forecast in Section 23.

13 Technical refinements and foundational inputs

Boundary heat-kernel coefficients (severity low). The Branson–Gilkey–Kirsten–Vassilevich boundary coefficients for the cap-shifted S^3 enter *only* the consistency checks, not the main derivation, which lives on the closed S^4 . The BFK gluing identity $\zeta_M(0; D^4(\theta_0)) + \zeta_M(0; D^4(\pi - \theta_0)) = 104/45$ holds for all θ_0 ; decomposing $\zeta_M = \zeta_{\text{bulk}} + \zeta_{\text{bnd}}$ and writing $\delta = \theta_0 - \pi/2$, the bulk part satisfies $\zeta_{\text{bulk}}(\pi/2 + \delta) + \zeta_{\text{bulk}}(\pi/2 - \delta) = -4a_\gamma$ and the boundary part $\zeta_{\text{bnd}}(\pi/2 + \delta) + \zeta_{\text{bnd}}(\pi/2 - \delta) = 3$ (both constant in δ). Hence $g(\delta) := \zeta_{\text{bnd}}(\pi/2 + \delta) - 3/2$ is an *odd* function: all even-order Taylor coefficients vanish identically. This is a rigorous theorem, not a perturbative estimate. The leading bulk shift is $|\Delta\zeta_{\text{bulk}}| = 3a_\gamma^2 \approx 0.088$; in the BFK-protected natural scheme the boundary shift is zero, and in the cap-scale scheme it is bounded by $|\Delta\Omega_K|/\Omega_K \lesssim 1.5\%$ (the L2 band of Table 1).

Scheme independence. The leading prediction $\mathcal{R} = a_\gamma/(8\pi^2)$ is built from the trace-anomaly integral (a renormalization-group invariant) and the Q -charge (a topological invariant); neither depends on the renormalization scale μ , so $\mathcal{R}$ is scheme-independent. The natural scheme ($\mu = R^{-1}$) is adopted throughout; the cap-scale scheme shifts only subleading corrections, by $\lesssim 1.5\%$.

Picard–Lefschetz contour analysis. The saddle equation $a_\gamma \cos^4 z = \sin z$ has one real saddle ($\delta\theta^* = 0.16391$, $\text{Im } S = 0$ exactly); every complex saddle in the box $|\text{Re } z| < 8$, $|\text{Im } z| < 2$ (sixteen of them) satisfies $|\text{Im } S| \geq 0.78$. By Picard–Lefschetz theory no Stokes line connects the real saddle to any complex saddle — the gap exceeds the connection threshold by a factor $> 7 \times 10^5$ — so the steepest-descent thimble through the cap is topologically isolated and $n_{\text{complex}} = 0$ is established *analytically* for the reduced cap-action, not merely numerically; the RK45 thimble flow confirms it independently. The cap saddle therefore dominates the reduced cap-action contour integral. The embedding of this result into the full infinite-dimensional no-boundary path integral is assembled semiclassically (Section 18): the cap-plus-scalar enumeration is complete, and the prediction $a_\gamma/(8\pi^2)$ is all-orders stable as a ratio of $\hbar$ -independent invariants. As a conditional fallback we retain the worst-case estimate — were a complex saddle nonetheless to contribute ($|n_{\text{complex}}| = 1$ in the full theory), the impact would be $|\epsilon_{\text{PL}}| \leq 6.6 \times 10^{-4}$ in Ω_K^{obs} — about 44% of the prediction, but below DESI DR2 sensitivity and within DESI DR3 / Euclid reach.

Foundational computations on the three residual points. Three explicit calculations sharpen the residual open items. *(i) The embedding B1'.* The Euclidean fluctuation Hessian was computed by shooting in all three transverse sectors (scalar, co-exact 1-form, transverse-traceless tensor) for $\ell \leq 12$ on the deformed cap; every eigenvalue is strictly positive (global minimum 8.16), and the centrifugal term $\ell(\ell+k)/\sin^2 \theta$ grows monotonically, so positivity for all higher ℓ is guaranteed asymptotically. The cap saddle is therefore a Morse-index-1 critical point (the single negative direction is the lapse), enlarging the transverse mode space adds only convergent Gaussian factors, and $n_{\text{complex}} = 0$ is preserved at one loop for any finite truncation. The genuine residual is non-perturbative (infinite-DoF, non-Gaussian), suppressed by $A_{\text{M5}} \approx 1.2 \times 10^{-6}$ — a limitation generic to all of semiclassical quantum gravity, not specific to this framework. *(ii) The unconditional sign.* The minisuperspace lapse on-shell action was derived symbolically for two boundary data: Dirichlet $q(0) = 0$ yields four saddles (including a complex pair with opposite-sign weights, the Feldbrugge–Lehners–Turok ambiguity), whereas Neumann $\dot{q}(0) = 0$ yields two real saddles — the analytic structure itself is boundary-condition-dependent, and the anomaly term ($a_\gamma/2 \approx 0.086$ against 6) does not lift it. The sign is thus fixed only once a no-boundary boundary condition (equivalently, an integration contour) is chosen; this is a foundational definitional input of quantum cosmology, not a CSG-KP-specific gap. *(iii) The reference epoch A5c.* The physical density ratio obeys $\Omega_K/\Omega_\Lambda = (\Omega_K^0/\Omega_\Lambda^0)(1+z)^2$ (since $\rho_K \propto a^{-2}$, $\rho_\Lambda = \text{const}$), ranging from 2.18×10^{-3} today to $\sim 2.6 \times 10^3$ at recombination. The prediction $a_\gamma/(8\pi^2)$ is specifically the $z = 0$ value; a reference epoch is therefore genuinely required and data-anchored, confirming that A5c is irreducible. The redshift-constancy established earlier is a property of the Euclidean conformal quantity, not of the physical density ratio.

The fixed-point value: reconciliation. Two legacy pipelines reported $\Lambda = -0.81331$ and $\Lambda^* \approx -0.691$. Both are *superseded* by the correction of Section 10: the $\langle R \rangle/4$ sharpness proxy on which they rested is a floored-geometry artefact (a constant $\Omega_\Lambda > 0$ does not recollapse, so the forced turnaround drove $H^2 < 0$ and was silently floored). The corrected Friedmann+Klein–Gordon dynamics give a *universal* fixed point Λ^* (independent of φ_0 and Ω_m) but only *amplitude-set*: $\Lambda^* \approx -0.10$ in $3H_0^2$ units for $A_{\text{CW}} = a_\gamma/128\pi^2$ (≈ -0.002 for the M5 amplitude), *not* the legacy -0.691 . What survives is structural and unchanged: Λ is not free, its sign is negative, the bulk vacuum cancels, and the iteration is a Banach contraction (constant $L \approx 0.18\text{--}0.30$,

sign $G' < 0$ analytic, $|G'| < 2$ numerical). What does *not* survive is a sharp absolute value: $\Omega_\Lambda = (V(\varphi_0) - \Lambda^*)/3$ leaves φ_0 free, so the homogeneous route fixes neither the sharp Λ^* nor the absolute Ω_Λ . None of this bears on the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$.

Conditional predictions. If the lightest neutrino is exactly massless, the Weyl anomaly $a_W = 11/720$ adds to the photon, giving $|\Omega_K| = 0.00162$ ($\Delta|\Omega_K| = 1.3 \times 10^{-4}$ from photon-only; Euclid+CMB-S4 discriminates at $\sim 0.5\sigma$). A second massless $U(1)$ would double the coefficient to $2a_\gamma = 62/180$. Both are falsifiable by DESI DR3 / Euclid.

Foundational inputs. The smooth Janus point at the equator depends on Axiom A1 (Hartle–Hawking no-boundary); the framework is compatible with alternative Janus-providing scenarios (Shape Dynamics, Boyle–Turok CPT-bounce) but A1 is the most-studied choice. The coefficient $a_\gamma = 31/180$ assumes the Bunch–Davies vacuum, the unique maximally symmetric Hadamard state on de Sitter; non-trivial alpha-vacua would modify it but are physically disfavoured without new structure.

14 Uniqueness: the classification and the falsified mechanisms

Proposition 3 (Uniqueness of the invariant). *Among dimensionless combinations of conformal-anomaly and Q -curvature data on D^4 , the ratio $a_\gamma/C_Q = a_\gamma/(8\pi^2)$ is the unique candidate satisfying: (i) scale invariance (no M_{Pl} , no G); (ii) conformal naturalness (Q_4 , not E_4 , as denominator); (iii) photon-only infrared content (Komargodski–Schwimmer selection); (iv) constancy of the action-parameter ratio $3|K|/\Lambda$ (epoch-independent as a ratio of fixed FRW constants, the density parameters redshifting as $1/a^2$).*

Rigorous core. The decisive structural fact is exact: on S^4 the Paneitz operator P_4 has kernel equal to the constants, so the constant conformal mode σ_0 is its *unique* zero mode, and $\int_{D^4} Q_4$ is the unique linear functional appearing as its Wess–Zumino source. Any alternative normalization either redefines σ_0 by a constant (leaving the ratio invariant) or is not the conjugate quantity; the Chang–Yang relation $\int Q_4/\int E_4 = 1/4$ provides an independent cross-check ($4a_\gamma/32\pi^2 = a_\gamma/8\pi^2$). *Honest limit.* Criteria (i)–(iv) are each implied by an axiom and are *sufficient* to select $8\pi^2$ from the natural candidate list; we do not claim they are minimal — a systematic proof that no broader admissible criterion set selects a different invariant is open, and the uniqueness is therefore established within the conjugate-pair plus (i)–(iv) framework, not as an unconditional theorem over all conceivable constructions.

A complementary constructive argument exhausts the dynamical alternatives: every attempt to obtain $\Omega_K^{\text{conformal}} = a_\gamma/(8\pi^2)$ from a *local field equation* rather than the topological quotient fails, organizing into six no-go categories — (I) conformal mechanisms linear in σ_0 or vanishing; (II) Lorentzian dynamics reintroducing Newton’s constant; (III) massive fields regenerating the 10^{120} hierarchy; (IV) squashing deformations only quadratic; (V) spectral invariants conformally protected; (VI) vacuum energy absorbed by sequestering. Table 2 lists the twenty-six explicit mechanisms.

Table 2: The twenty-six falsified mechanisms for deriving $\Omega_K^{\text{conformal}} = a_\gamma/(8\pi^2)$ from local dynamics.

#	Mechanism	No-Go	Failure mode
1	Riegert variational	I	linear in σ_0 ; no σ_0^2
2	DtN determinant (naive)	I	too weak by 10^{240}
3	KP stiffness	I	conformal sector unchanged

#	Mechanism	No-Go	Failure mode
4	Gauss–Bonnet (conformal)	I	identically zero on FRW
5	Gauss–Bonnet (geometric)	I	cubic in a ; no σ_0^2
6	Resonance $4Q_4 = \mu_1$	I	coupling coefficient ≈ 0
7	Möbius nonlinearity	I	no 4D solution
8	Topology sum	I	not regularizable
9	Conformal Perelman flow	I	P_4 remains covariant
10	Two-loop stiffness	I	zero for free conformal fields
11	$L = 1$ coupled system	I	σ_0^* is a_γ -independent
12	Counter-field	I	symmetric hemispheres cancel
13	C_Q -decoherence	II	requires $K = C_Q^2$ input
14	Q -prescription	II	restoring force = 0 after KP
15	Topology path integral	I	zero-mode divergence
16	Fokker–Planck for σ_0	II	gives $\Omega_K \propto 1/a_\gamma$ (wrong)
17	Lorentzian field equation	II	$(H/M_{\text{Pl}})^2$ suppression
18	Anomalous field equation	IV	squashing only quadratic
19	Conformal scalar VEV	III	$V(\sigma_0) = \text{const}$
20	Coleman–Weinberg	I	β -function = anomaly
21	Explicit mass m^2	III	hierarchy of 10^{249}
22	Topological mass ξR	III	wrong sign
23	$\zeta'(0)$ spectral	V	transcendental constant
24	Lichnerowicz spectral	V	no σ_0 dependence
25	Heat-kernel spectral	V	rigid on S^4
26	Stochastic	VI	$\langle C^2 \rangle$ absorbed by KP

The lesson is methodological: the correct route is not a new field equation but the topological hemisphere quotient, with the cap-saddle providing the geometric realization.

15 Historical methodology

Five reference cases share one logical pattern, *compact space + local condition $\Rightarrow$ global parameter*: Balmer→Bohr 1913 (electron on a compact configuration space, single-valuedness selects discrete levels); Dirac 1931 ($eg = n\hbar/2$ from S^1 single-valuedness around the string); the quantum Hall effect 1980 ($\sigma_{xy} = ne^2/h$ from the integer Chern number on a torus); Hawking–Gibbons 1977 ($T = \kappa/2\pi$ from regularity of the compact Euclidean section); and Casimir 1948 ($\rho = -\pi^2/720d^4$ from boundary conditions on a compact transverse interval). CSG–KP has the compact space (D^4) and the conditions (KP sequestering, photon-uniqueness, Janus matching) but, by the 26-mechanism exhaustion (Section 14), no *local* mechanism that singles out a value: there is no Hawking conical regularity (the equatorial S^3 is maximal, not conical), no Dirac quantization (the conformal connection is trivial), no Hall index (η is conformally invariant), no Casimir extremum, and no anomaly-matching obstruction (the FLRW slice is conformally flat throughout). The prediction therefore belongs to a distinct category — a topological/anomalic ratio, not a regularity-quantized observable — which is the structural content of the complementarity argument.

16 The coincidence problem: structure-formation averaging

The third face of the cosmological constant problem — why Ω_Λ and Ω_m are comparable now — is not solved by the ratio (1), whose constant object is the action-parameter ratio $\mathcal{R} = 3|K|/\Lambda$ (the density-parameter ratio itself redshifts as $\mathcal{R}/a^2$). A clean physical constraint narrows the solution space: a dynamical dark-energy component cannot *simultaneously* have $w \approx -1$ (data) and track matter (coincidence); the two are mutually exclusive, and sequestering, being a constant subtraction, leaves $w_{\text{eff}} = -1$. The coincidence is therefore resolved not by dark-energy *dynamics* but by the *averaging prescription* for Λ , following Lombriser’s structure-formation counterpart to Kaloper–Padilla [20].

Lombriser’s mechanism replaces the homogeneous four-volume average of Section 10 by an average weighted toward gravitationally bound (halo) regions, evaluated with the spherical-collapse top-hat evolution

$$y'' + \left(2 + \frac{H'}{H}\right)y' + \frac{1}{2}\Omega_m(a)(y^{-3} - 1)y = 0, \quad ' \equiv \frac{d}{d \ln a}, \quad (20)$$

with $y = (\rho_m/\bar{\rho}_m)^{-1/3}$ and matter-dominated initial data $y_i = 1 - \delta_i/3$, $y'_i = -\delta_i/3$. The single additional assumption is a uniform prior on y , i.e. $y(t_0) = \frac{1}{2}$. With the equality normalization ($a_{\text{eq}} \equiv 1$, $\Omega_m(a) = a^{-3}/(a^{-3} + 1)$) the critical (longest-lived) shell has overdensity $\delta_i^{\text{crit}} = 1.134 \times 10^{-3}$ and reaches $y = \frac{1}{2}$ at $a_0/a_{\text{eq}} = 1.336$, giving

$$\Omega_\Lambda = \frac{1}{a_0^{-3} + 1} = 0.7044. \quad (21)$$

This value is *derived*, not calibrated — but conditional on the uniform-prior assumption $y(t_0) = \frac{1}{2}$, which is anthropic-like and is not itself derived from the framework. The homogeneous and halo-weighted averages are two faces of the same $\Lambda = \frac{1}{4}\langle R \rangle$ structure, differing in averaging prescription. The *homogeneous* (volume) route delivers, as corrected in Section 10, the universality *structure* of Λ^* and the order of magnitude only — not the absolute scale, since φ_0 remains free. The *halo-weighted* route returns 0.704 but at the cost of the anthropic-like $y(t_0) = \frac{1}{2}$. The Euclidean topology fixes the dimensionless *ratio* Ω_K/Ω_Λ rigorously; the absolute *scale* is fixed by neither the volume route (free φ_0) nor without a prior by the halo route. The resulting trio,

$$\Omega_\Lambda = 0.7044, \quad \Omega_K = \frac{a_\gamma}{8\pi^2} \Omega_\Lambda = 1.54 \times 10^{-3}, \quad \Omega_m = 0.2956, \quad (22)$$

is internally consistent and within 2.8% (Ω_Λ) and 0.7σ (Ω_K) of observation.

16.1 A geometric budget ladder (conjecture)

A second, sharper route to the absolute scale is suggested by the numbers themselves, and is recorded here *as a conjecture*, with its open step made explicit. The factors that appear across the framework form a $2^n\pi^2$ ladder of 4D conformal geometry: the one-loop factor $16\pi^2 = 2^4\pi^2$ in $\langle T \rangle = a_\gamma E_4/16\pi^2$; the hemisphere Euler integral $\int_{D^4} E_4 = 32\pi^2 = 2^5\pi^2$ ($\chi = 1$); and the full-sphere integral $\int_{S^4} E_4 = 64\pi^2 = 2^6\pi^2$ ($\chi = 2$). Within this ladder the three budget quantities all follow from a_γ and the Euler integrals:

$$\Omega_\Lambda = \int_{S^4} \langle T \rangle \sqrt{g} = 4a_\gamma = 0.6889, \quad (23)$$

$$\frac{\Omega_K}{\Omega_\Lambda} = \frac{4a_\gamma}{\int_{D^4} E_4} = \frac{a_\gamma}{8\pi^2} = 2.181 \times 10^{-3}, \quad (24)$$

$$\Omega_K = \frac{\Omega_\Lambda^2}{\int_{D^4} E_4} = \frac{a_\gamma^2}{2\pi^2} = 1.503 \times 10^{-3}. \quad (25)$$

The same numerator $4a_\gamma$ appears in the leading term and in the ratio; dividing by $\int_{D^4} E_4 = 32\pi^2$ turns the full-sphere anomaly into the cap ratio. This yields the exact algebraic identity $\Omega_K \cdot 32\pi^2 = \Omega_\Lambda^2$, and a *perturbation hierarchy in a_γ* : dark energy is the leading anomaly effect ($\Omega_\Lambda \sim a_\gamma^1$) while curvature is second order ($\Omega_K \sim a_\gamma^2$). The smallness $|\Omega_K| \ll \Omega_\Lambda$ is then structural — curvature is quadratically suppressed in the anomaly — not a tuning, mirroring the holographic $V_4 = A_2^2$ relation of Section 19. The split is geometric: the full S^4 (the timeless no-boundary substrate) carries the absolute Ω_Λ , the hemisphere D^4 (the observer past light cone) carries the ratio, and $32\pi^2 = \int_{D^4} E_4$ is the substrate-to-observer bridge. The full budget $\Omega_\Lambda = 0.6889$, $\Omega_m = 0.3096$, $|\Omega_K| = 1.5 \times 10^{-3}$ matches Planck 2018 (base Λ CDM) at 0.0σ (Ω_Λ), -0.27σ (Ω_m), consistent (Ω_K), from $a_\gamma = 31/180$ and $\chi = 2$ alone.

Status (the one open step). The ladder *algebra* is exact and the factor $32\pi^2$ is geometric, not a free normalization; the ratio $a_\gamma/8\pi^2$ is theorem-level. What is *not* derived is the identification (23) itself: why the dark-energy density *fraction* should equal the integrated anomaly *action* on the full sphere. The local energy-density route ($\rho_{\text{anom}}/\rho_{\text{crit}} \times N_{\text{eff}}$, the de Sitter identity of Section 19) yields only the *ratio* $a_\gamma/8\pi^2$, not $4a_\gamma$; an action/topological identification of the dark energy is required, which A5 does not supply (A5 establishes the ratio only), and which is in tension with the KP sequestering of the Λ -channel. The numerical match is therefore striking and the structure coherent, but (23) is a conjecture pending this action-identification bridge — not a proven prediction of the absolute scale.

Refinement (this work): the prefactor is computed, the residue localized, the scope confined. Three results sharpen the open step without closing it. (i) *The lift prefactor is computed.* The factor identification (23) requires, $\rho_\Lambda/\rho_{\text{anom}} = 32\pi^2(M_{\text{Pl}}/H)^2$, is exactly the Bekenstein–Hawking horizon degree-of-freedom count $A/G = 4S_{\text{dS}}$ — a standard holographic quantity, not a free normalization — and with it $\Omega_\Lambda = (A/G)\rho_{\text{anom}}/\rho_{\text{crit}} = 4a_\gamma$ falls out, the coherent ($\times N$) accumulation being the unique scale-free choice (incoherent $\sqrt{N}$ gives an H -dependent value). (ii) *The residual assumption is localized.* An in-in (Schwinger–Keldysh) evaluation shows the lift is not an in-in mode sum (that is already inside ρ_{anom}) but the Cohen–Kaplan–Nelson UV–IR bound $\rho_\Lambda \leq M_{\text{Pl}}^2/L^2$: its Schwarzschild-consistency part *follows* from gravity and fixes the 10^{-122} magnitude, while its *saturation* is the assumption P5 — now a named holographic principle, not a vague accumulation. (iii) *The scope is confined.* This residue sits in the absolute value alone: the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ is scale-invariant (H and M_{Pl} cancel identically), so the falsifiable Euclid prediction is independent of the unproven saturation; deriving the saturation from scale-invariance is a fallacy, since scale-invariance constrains only the ratio. The conjecture is thereby sharpened — prefactor computed, principle named (CKN saturation), scope limited to the conjectural bonus — but not proven.

A no-go: the value is an action invariant, not a dynamical equilibrium. The open step above is structural, not a missing calculation. Every dynamical or energy-density route to the absolute Ω_Λ gives a *different* number, and none gives $4a_\gamma$:

- KP sequestering $\rho_\Lambda = \int \langle T \rangle / V_4 - (a_\gamma/8\pi^2)(H/M_{\text{Pl}})^2 \sim 10^{-122}$;
- the canonical Euclidean vacuum energy $\rho_\Lambda = -dS_E/dV_4$ on the S^4 saddle — $\frac{1}{2}$ (the gravitational term dominates; the anomaly enters at 10^{-122});
- the M5/KP fixed point $1/12a_\gamma \approx 0.484$; the Coleman–Weinberg iteration (φ_0 free); the Riegert mode (linear, no saddle);
- the causal closures over the forced apparent-horizon surface — bare dimensional choices with median ≈ 0.93 , KP stationarity $\langle R \rangle_M/4 \approx 14$, and the Lorentzian light-cone anomaly too weak by $(M_{\text{Pl}}/H)^4 \sim 10^{240}$.

The *only* route to $4a_\gamma$ is the pure action invariant $\int_{S^4} \langle T \rangle \sqrt{g} = 4a_\gamma$ ($= |\zeta(0)|$ = the Riegert scale-flow), cutoff-free and exact but an *action*, not a dynamically forced density. Hence $4a_\gamma$ is not a dynamical equilibrium value at all; it is a topological-spectral invariant, and the identification of the dark-energy fraction with it cannot be derived from any dynamics — each dynamics demonstrably yields something else. This is the precise structural reason the step is open.

The sharpest form of the conjecture. The identification $\Omega_\Lambda = 4a_\gamma$ is exactly equivalent to a single normalization,

$$N_{\text{eff}} = \frac{S_{dS}}{C_Q} = \frac{8\pi^2 M_{\text{Pl}}^2 / H^2}{8\pi^2} = \left(\frac{M_{\text{Pl}}}{H} \right)^2, \quad (26)$$

the Gibbons–Hawking de Sitter entropy per Q -curvature charge. With this $c = 1$ normalization the coherent lift of Section 10.4 returns $\Omega_\Lambda = 4c a_\gamma = 4a_\gamma$. The open question thereby narrows from “a vague action identification” to “why this one geometrically natural normalization (entropy per Q -charge)”. It is sharper and more falsifiable, but S_{dS}/C_Q is not more forced than the bulk-projection prefactor $4/3\pi$, so it remains a conjecture — now in its tightest form.

16.2 Why the sharp value is not pinned by the history ensemble: the measure problem

The conditional status of $\Omega_\Lambda = 0.704$ deserves a sharper statement, because two independent attempts to remove the uniform-prior assumption were carried through and both fail for the same categorical reason. First, Lombriser and CSG–KP are the *same constraint class*: Lombriser varies the Einstein–Hilbert action with respect to the Planck mass, producing a global Lagrange-multiplier constraint on the averaged Ricci scalar $\langle R \rangle$ — structurally identical to $\Lambda^* = \frac{1}{4} \langle R \rangle_{\text{ren}}$ of Proposition 2. The single free element is the prior on the observer’s position in structure formation.

The collapse-time lever. The value 0.704 is the limit of *infinitely long-lived* shells ($a_{\text{collapse}} = 6686$ at equality normalization); for a finite maximum shell age Ω_Λ decreases, and the observed $\Omega_\Lambda = 0.685$ is reached already at $a_{\text{collapse}} \approx 8.67$. CSG–KP appears to supply exactly what Lombriser lacks: Proposition 2 forbids $t_{\text{max}} \rightarrow \infty$, since a finite four-volume requires a background recollapse driven by the Coleman–Weinberg potential, which would truncate the shell lifetime and pull 0.704 toward observation. *The lever fails quantitatively.* Forward integration of the full background dynamics (Friedmann + Klein–Gordon with $V(\varphi) = A\varphi^4(2\ln(\varphi/v_S) - \frac{25}{6})$, $A \approx 3.6 \times 10^{-6}$, $\varphi_0 \approx 22.9 v_S$, $\varphi_{\text{crit}} \approx 8.0 v_S$, $m_\varphi \approx 0.27 H_0$) gives recollapse at $a_{\text{max}} \approx 8 \times 10^{17}$ (~ 41 e -folds, $t \sim 170/H_0$), whereas $\Omega_\Lambda = 0.685$ would require $a_{\text{max}} \approx 6.7$ — seventeen orders of magnitude too late. The light scalaron rolls too slowly from $\varphi_0 \gg \varphi_{\text{crit}}$, so the recollapse lies far beyond observational horizons (the explicit caveat of Section 10) and cannot truncate the shell lifetime in the relevant range.

The measure lever. A second attempt replaces the uniform-in- y prior by the Hartle–Hawking amplitude $|\Psi(\sigma_0)|^2 \propto e^{-8a_\gamma \sigma_0}$, the canonical no-boundary weight over cosmological histories. The amplitude alone does not fix the observer weighting; combined with different companion measures it ranges over the entire admissible interval — $\text{HH} \times dN \rightarrow 0.00$, $\text{HH} \times dt \rightarrow 0.04$, $\text{HH} \times V_4 \rightarrow 0.78$, $\text{HH} \times (\text{phase space}) \rightarrow 1.00$. The framework’s own canonical choice (the amplitude *is* the V_4 weighting) gives 0.78, not 0.685. The companion measure one would need is physically the observer density — when and where observers exist — which no axiom fixes. This is the *anthropic* measure problem of quantum cosmology, and in that form it is categorical, not technical: weighting a history ensemble by an observer density that the theory does not supply, Ω_Λ cannot be pinned by any prior-independent rule.

The causal route avoids the measure problem entirely. The anthropic difficulty arises only because one weights an *ensemble* of histories. A causal variational closure removes the ensemble altogether: fixing Λ by stationarity over a single past light cone (Section 9) needs no observer measure. The self-consistency $\Lambda = 3\Omega_\Lambda H_0^2 \sim c/t_U^2$ then has a prior-free solution depending only on the dimensionless light-cone coefficient $c = A_{\partial M}/(t_U^{-1}V_M) = O(1)$, where $A_{\partial M}$ is the boundary 3-volume and V_M the four-volume of the past light cone. Two questions must be separated: which boundary hypersurface, and which functional form.

The hypersurface is forced. It is not a free choice: the maximal-enclosed-3-volume slice (the “belly”) coincides *exactly* with the apparent horizon, the covariant $\theta = 0$ surface, since $\dot{r} = Hr - 1 = 0 \Leftrightarrow r = 1/H$ for the proper light-cone radius $r = a\chi_{lc}$. Two independent characterizations — maximal 3-volume and the covariant Bousso/Hayward $\theta = 0$ surface — select the same slice (numerically $rH = 1.000$). The particle horizon, a candidate alternative, is the big-bang tip with $rH \approx 55$ and is *not* a covariant causal boundary; it is excluded. The apparent-horizon choice is therefore forced, and the earlier impression of a $\sim \times 500$ “definition dependence” was an artifact of comparing the genuine causal boundary with a non-boundary.

The functional form is not forced. With the unique apparent-horizon surface, no available principle fixes the dimensionless closure. The natural candidate to *derive* it — KP stationarity, $\Lambda = \frac{1}{4}\langle R \rangle$ over the averaging 4-volume, here restricted to the causal diamond, $\Lambda = \frac{1}{4}\langle R \rangle_M$ over V_M — *fails*: the past-cone average is dominated by the early-matter a^{-3} curvature, giving $\langle R \rangle_M/4 \sim 14 \gg 3\Omega_\Lambda$ with no self-consistent fixed point. Apparent-horizon thermodynamics do not help either: the first law at the horizon (Cai–Kim) reproduces the Friedmann equation identically and imposes no Λ constraint, while de Sitter entropy saturation gives the trivial $\Omega_\Lambda = 1$. And the bare dimensional closures spread across almost the whole admissible range: over the unique apparent-horizon surface, natural choices give $\Omega_\Lambda \in \{0.489, 0.725, 0.929, 0.938, 0.959\}$ (median 0.93), with no cluster near the observed value. The often-quoted $3\Omega_\Lambda = V_3(\text{AH})/(V_M t_U) \rightarrow 0.725$ is therefore *one selective choice*, not a distinguished one — most natural closures land near 0.93. (The earlier band value $c = 1.86$ was *circular*, choosing $L = t_U$ and reading $c = \Lambda t_U^2$ off the input, and is superseded; Shaw–Barrow’s sharp $\Omega_K = -0.0056$ likewise needs extra physical inputs, the QCD condensate $\zeta_b \approx \frac{1}{2}$ and the inflationary e -fold count N , not pure geometry.) So the causal route does *not* predict 0.685, nor even a narrow band: it forces the hypersurface (apparent horizon) and an $O(1)$ range, but the closure functional is undetermined and the obvious candidates (KP stationarity, horizon thermodynamics) are ruled out. The open question is thereby sharpened — from “a free prior” through “which boundary” (answered: the apparent horizon) to “which closure functional,” which remains genuinely open. What is fixed prior-free is therefore: the dimensionless ratio $\Omega_K/\Omega_\Lambda = a_\gamma/(8\pi^2)$ (theorem), the magnitude and coincidence to order unity, the forced apparent-horizon hypersurface, and an $O(1)$ range for Ω_Λ — not the sharp value, which is pending a derived closure functional, not a free measure.

17 The sign: anomaly-forced curvature and the Janus point

The magnitude of Ω_K follows from Eq. (1); the sign is fixed by the cap action itself. Integrating the stationarity condition $d\Gamma/d\theta_0 = 3a_\gamma \sin^3 \theta_0 + 3 \cot \theta_0$ gives the reduced cap action $\Gamma(\delta\theta) = 3a_\gamma \sin \delta\theta - a_\gamma \sin^3 \delta\theta + 3 \ln \cos \delta\theta$, whose only real stationary point in $(-\frac{\pi}{2}, \frac{\pi}{2})$ is $\delta\theta^* = +0.16391$ for the physical $a_\gamma = +31/180 > 0$ — an *open* universe ($\Omega_K > 0$). There is no closed-universe saddle: a negative $\delta\theta^*$ would require $a_\gamma < 0$. The saddle *location* depends only on the action, whose anomaly source $+a_\gamma$ is the prescription-independent photon heat-kernel coefficient; the Hartle–Hawking versus Vilenkin choice ($\Psi \propto e^{-S_E}$ versus e^{+S_E}) reweights this same saddle but does not move it. The sign is therefore forced by $\text{sign}(a_\gamma) > 0$ together with the dominance of the real cap saddle (B1, $n_{\text{complex}} = 0$), not by a branch postulate. The Wick rotation is the established Bucher–Goldhaber–Turok open-inflation construction [16, 17]: the contour $\theta =$

$\pi/2 + i\tau$ carries the compact cap into a hyperbolic foliation with the conical singularity off the integration contour, so the smooth-beginning obstruction of [18] does not apply. The Janus point — the moment of vanishing extrinsic curvature [19] — is the equatorial S^3 where the two hemispheres meet under a Z_2 reflection.

Status of the sign. The curvature sign separates into two distinct sectors. The *curvature sector* — whether the cap deforms toward an open or a closed continuation — is fixed: the cap action has a unique real saddle at $\delta\theta^* > 0$ for $a_\gamma > 0$, and the explicit computation (Section 18) confirms its location is independent of the $e^{\mp S_E}$ weighting, so the open sign is forced by $\text{sign}(a_\gamma) > 0$ and the real-saddle dominance B1, modulo the prescription-independence of the anomaly-induced action (standard for a matter heat-kernel coefficient). The *fluctuation-stability sector* — whether the Lorentzian continuation of the lapse integral is well defined and normalizable — is the Feldbrugge–Lehners–Turok question: the explicit lapse-saddle count is boundary-condition dependent (four saddles under Dirichlet data, two under Neumann), so the well-definedness of the Lorentzian no-boundary measure is a foundational input of quantum cosmology, shared by every no-boundary model and not specific to this framework. The two sectors are independent: the FLT ambiguity concerns the stability of fluctuations, not the sign of the spatial curvature, which is already fixed in the Euclidean cap by $a_\gamma > 0$. Empirically, DESI DR2 places the open prediction at 0.74σ and disfavors the closed (Vilenkin) reading at 4.6σ .

18 Foundational computations: sign, $B1'$, and the reference epoch

This section records explicit computations targeting the three items the framework had carried as non-theorem-level. The outcome is mixed and stated without embellishment: one item is closed, one is reclassified and closed in its physical part, and one is confirmed irreducible.

Curvature sign (closed). The reduced cap action $\Gamma(\delta\theta) = 3a_\gamma \sin \delta\theta - a_\gamma \sin^3 \delta\theta + 3 \ln \cos \delta\theta$, obtained by integrating the stationarity condition of Section 17, has exactly one real stationary point in $(-\frac{\pi}{2}, \frac{\pi}{2})$: $\delta\theta^* = +0.16391$ for $a_\gamma = +31/180$, with $\Gamma(\delta\theta^*) = 0.0431$. For $a_\gamma < 0$ the unique saddle moves to $\delta\theta^* = -0.16391$. Because the saddle location is determined by Γ alone — whose anomaly source $+a_\gamma$ is the prescription-independent photon heat-kernel coefficient — the Hartle–Hawking versus Vilenkin choice (e^{-S_E} versus e^{+S_E}) reweights but does not relocate it. Hence the open sign is forced by $\text{sign}(a_\gamma) > 0$ and the real-saddle dominance B1, not by a branch postulate.

$B1'$: the scalar embedding (cap dominance preserved). Two layers establish that the cap-saddle dominance survives the enlargement beyond the one-dimensional reduced action. First, at one loop the deformed cap $D^4(\theta_0 = \pi/2 + \delta\theta^*)$ is a polar cap on S^4 with S^3 slices, so $SO(4)$ is unbroken, ℓ remains a good quantum number, and the fluctuation Hessian is exactly block-diagonal in (ℓ, sector) with no ℓ – ℓ' mixing; its positivity is established analytically by domain monotonicity (Section 12), so the cap is a Morse-index-1 point and $n_{\text{complex}} = 0$ is preserved exactly.

Second — the perturbative embedding proper — the scalar matter field φ is restored as an independent degree of freedom, giving the two-dimensional action $\Gamma_{2D}(z, \varphi) = \Gamma_{\text{geom}}(z) - \frac{1}{2}\mu_{\text{eff}}^2\varphi^2 V_{\text{cap}}(z)$ with $\Gamma_{\text{geom}}(z) = 3a_\gamma(\sin z - \frac{1}{3}\sin^3 z) + 3 \ln \cos z$ (identical to the cap action of Section 17) and $V_{\text{cap}}(z) = \int_0^z \sin^3 t dt$. The 1D saddles lift trivially: since the Coleman–Weinberg potential satisfies $V_{\text{CW}}(0) = 0$, one has $\Gamma_{2D}(z, 0) = \Gamma_{\text{geom}}(z) = \Gamma_{1D}(z)$ exactly, so the eight 1D saddles are also 2D saddles with $\text{Im} \Gamma$ unchanged (Lemma B1-Stokes inherits). The scalaron-vacuum saddle at $\varphi = v_S e^{11/6}$ lies only $\sim A_{M5} \approx 1.2 \times 10^{-6}$ below the geometric cap action (a relative shift $\sim 3 \times 10^{-5}$ of $\Gamma_{\text{geom}} \approx 0.043$), and the single additional 2D-specific saddle, at $z \approx -0.656 + 1.241i$, has $\text{Im} \Gamma \approx 1.87 \neq 0$ — no Stokes line connects it to the real cap. Cap

dominance is therefore preserved under the scalar perturbation, with the correction suppressed by A_{M5} .

Complete thimble enumeration. The enumeration can be closed rigorously rather than by sampling. Substituting $u = e^{iz}$ turns the saddle equation $a_\gamma \cos^4 z = \sin z$ into the polynomial $a_\gamma(u^2 + 1)^4 + 8i(u^5 - u^3) = 0$, of degree 8; by the fundamental theorem of algebra this returns *all* eight saddles exactly. One is the real cap ($z^* = 0.1639$, $\text{Im } \Gamma = 0$); the other seven are complex with $|\text{Im } \Gamma| \geq 1.87 > 0$. On the scalaron branch $\varphi = v_S e^{11/6}$ the φ -equation decouples and the z -equation becomes the κ -perturbed degree-8 polynomial ($\kappa = \mu_{\text{eff}}^2 \varphi^2 / 6 \approx 2.4 \times 10^{-3}$), again exactly eight saddles — one near-real (the scalaron, at $\text{Re } \Gamma$ lower than the cap by $\sim A_{M5}$) and seven complex with $|\text{Im } \Gamma| > 0$. The decisive step is Picard–Lefschetz: along any Lefschetz thimble $\text{Im } \Gamma$ is constant. The real no-boundary contour carries $\text{Im } \Gamma = 0$, while every one of the fourteen complex saddles has $\text{Im } \Gamma \neq 0$, so each complex thimble keeps $\text{Im } \Gamma = \text{const} \neq 0$ and never reaches the real axis. Hence $n_{\text{complex}} = 0$ *rigorously* — the conclusion follows from the constancy of $\text{Im } \Gamma$ along thimbles, not from a finite numerical saddle scan. The dominant contribution is the pair of near-degenerate real saddles (cap and scalaron, separated by $\sim A_{M5} \sim 10^{-6}$), and the cap sector dominates the no-boundary wave function. The only ingredient outside the polynomial closure is the set of complex log-branches of $V'(\varphi) = 0$ ($\varphi^2 = v_S^2 e^{11/3 + 2\pi i n}$): these are renormalization artifacts of the Coleman–Weinberg logarithm rather than saddles of the fundamental theory, and are in any case exponentially suppressed, so they do not affect $n_{\text{complex}} = 0$.

Embedding into the full no-boundary path integral (semiclassical). The pieces above assemble into a complete semiclassical evaluation of $\Psi_{\text{NB}} = \int \mathcal{D}g \mathcal{D}\varphi e^{-S_E[g, \varphi]} = \sum_a n_a e^{-\Gamma_a} [\det' M_a]^{-1/2} (1 + O(\hbar))$. (i) The dominant saddle is the maximally symmetric cap $D^4 \subset S^4$, so the saddle search reduces to the homogeneous mini-superspace (z, φ) whose sixteen saddles are completely enumerated above; Picard–Lefschetz gives $n_{\text{complex}} = 0$, leaving the two real saddles (cap and scalaron, gap $\sim A_{M5}$) with the cap dominant. (ii) The inhomogeneous ($\ell \geq 1$) modes are Gaussian about the cap with a positive-definite Hessian (domain monotonicity, Section 12); their ζ -regularized determinant carries the scale anomaly $\zeta(0; \text{Maxwell}; S^4) = -\frac{31}{45} = -4a_\gamma$, so the one-loop determinant $[\det' M]^{-1/2}$ is itself the carrier of the prediction $a_\gamma / (8\pi^2)$. The inhomogeneous modes have vanishing overlap with the anomaly source on D^4 by parity (the $\ell = 1$ S^3 harmonic integrates to zero against E_4), so the factorization homogeneous $\times$ inhomogeneous is exact at quadratic order. (iii) By the Adler–Bardeen theorem the Type-A coefficient a_γ receives no higher-loop corrections, so the leading prediction is protected; the residual higher-loop shift is of order $A_{M5} \sim 5 \times 10^{-4}$ relatively, below DESI DR2 sensitivity. (iv) The single $\ell = 0$ conformal/lapse mode is rotated to convergence by Gibbons–Hawking–Perry. The honest limit is precise: this is *one-loop exact* for the Type-A prediction, not an all-orders non-perturbative evaluation. Full convergence of the Euclidean quantum-gravity path integral remains a generic open problem (the conformal-factor issue, mitigated but not resolved by GHP), shared with every approach to semiclassical quantum gravity and not claimed here.

All-orders stability of the prediction. The convergence question splits cleanly, and the physically relevant half is provable. The observable is the ratio $R = a_\gamma / (8\pi^2)$, and R is the quotient of two $\hbar$ -independent quantities, so it receives no perturbative correction at any loop order. First, CSG treats gravity classically: graviton loops do not exist, so the genuine obstruction to convergence of a Euclidean gravity path integral — the unboundedness of the conformal-factor action and the perturbative non-renormalizability of Einstein gravity, both graviton-loop effects — is structurally absent, and the remaining integral is matter on a classical saddle. Second, on the round S^4 the Weyl tensor vanishes ($W^2 = 0$), so Type-B drops out; the surviving Type-A coefficient $a_\gamma = 31/180$ is Adler–Bardeen one-loop exact, receiving no higher-loop correction. Third, $8\pi^2 = \int_{D^4} Q_4$ is a topological charge, a discrete/rational invariant independent of $\hbar$ to all orders. Fourth, the saddle sum is finite-dimensional with $n_{\text{complex}} = 0$. Hence R is all-orders

well-defined within CSG. The honest boundary is equally precise: the absolute Borel summability of the four-dimensional φ^4 matter series (the generic triviality/Landau question) and the normalization of $|\Psi|^2$ are *not* settled here, but both cancel from the ratio and are protected by its topological character — they do not bear on R . This is not a solution of the general quantum-gravity path integral; it is the decoupling of the prediction from that open problem, and it holds within the CSG assumption of classical/conformal gravity (itself an empirical question).

Fluctuation stability in the Euclidean sector (resolved). The single negative direction of the cap saddle is the conformal/scale-factor mode: $d^2\Gamma/d\delta\theta^2(\delta\theta^*) = -3.33 < 0$, while all transverse modes are positive (B1' above). This is the Euclidean Gibbons–Hawking–Perry conformal-factor situation, and its standard resolution applies: rotating the conformal mode along the imaginary axis, $\delta\theta = \delta\theta^* + is$, makes $\text{Re } \Gamma$ grow ($0.043 \rightarrow 0.060 \rightarrow 0.109$ for $s = 0, 0.1, 0.2$), so $e^{-\Gamma}$ decays and the Gaussian over the rotated mode converges. The Euclidean one-loop no-boundary wavefunction is therefore well defined. The Feldbrugge–Lehners–Turok obstruction concerns the *Lorentzian* lapse integral — a different definition of the path integral, which this framework does not adopt; the Euclidean cap is its primary object. Whether the Euclidean (GHP-rotated) or the Lorentzian definition is the physically correct quantum gravity is an open foundational debate, not a CSG–KP-specific gap.

The reference epoch A5c (no fit freedom, identification irreducible). The physical density ratio $\Omega_K(z)/\Omega_\Lambda(z) \propto (1+z)^2$ is epoch-dependent (today 2.18×10^{-3} , at $z = 1100$ about 2.6×10^3), so a reference epoch must be specified. It introduces no fit freedom: given the measured Ω_Λ , the prediction $\Omega_K = (a_\gamma/8\pi^2)\Omega_\Lambda$ is unique, with no tunable parameter. A5c is the identification rule “the Euclidean topological number equals the observable at the observation epoch,” forced by testability (we measure at $z = 0$). Anchoring instead at asymptotic de Sitter ($\Omega_\Lambda \rightarrow 1$) would shift the predicted Ω_K to $a_\gamma/8\pi^2$, a $\leq 46\%$ change; that the matching surface is the present epoch rather than the asymptotic future is a physical identification, not derivable from first principles.

Status after three iterations. Under the framework’s own (Euclidean, observation-anchored) definitions, every item closes: the curvature sign by $a_\gamma > 0$, B1' at one loop ($SO(4)$ -exact), Euclidean fluctuation stability by GHP rotation plus transverse positivity, and A5c without fit freedom. The two genuinely irreducible residues are interpretational rather than computational, and shared with the wider field rather than specific to this framework: the choice between a Euclidean and a Lorentzian path-integral definition, and the physical identification A5 of the Euclidean invariant with the observed curvature. These are not gaps to be computed away; they are the irreducible interpretational content, and observation rather than further algebra is what bears on them.

19 Physical interpretation: the Photon-Field Substrate

The mathematical structure admits a physical reading, the Photon-Field Substrate (PSF), built on five postulates: universal vacuum entanglement; the photon as universal massless carrier; the conformal anomaly as the substrate’s residual stress; the Komargodski–Schwimmer a -theorem as the infrared selection principle; and holographic accumulation $N_{\text{eff}} = M_{\text{Pl}}^2/H^2$. The number $a_\gamma/(8\pi^2)$ then emerges from two independent identities — the de Sitter structural identity $\rho_{\text{anom}}/(3H^4) = a_\gamma/(8\pi^2)$ and the hemisphere topological quotient $\Delta\Gamma/C_Q = a_\gamma/(8\pi^2)$ — joined by two physical identifications (the holographic Friedmann density and the curvature observable A5). PSF is complementary to the CSG–KP mathematical structure in the Bohr sense [24], and is consistent with locality and with loophole-free Bell tests [26, 27, 28], the correlations being kinematic vacuum entanglement (Reeh–Schlieder [25]) rather than signalling.

The magnitude hierarchy is carried by P5, and P5 is a postulate. The topological ratio $a_\gamma/(8\pi^2) \approx 2.2 \times 10^{-3}$ is dimensionless and clean, but its physical realization carries the full cosmological-constant hierarchy. A single-mode anomaly inserted naively into the Friedmann balance gives

$$\Omega_{\text{anom}}^{\text{naive}} = \frac{a_\gamma}{8\pi^2} \cdot \frac{H^2}{M_{\text{Pl}}^2} \sim 10^{-3} \cdot 10^{-122} \sim 10^{-125}, \quad (27)$$

122 orders of magnitude below the observed curvature density. The holographic accumulation postulate P5, $N_{\text{eff}} = M_{\text{Pl}}^2/H^2$, compensates this exactly: $\rho_{\text{eff}} = N_{\text{eff}} \rho_{\text{anom}} = (3a_\gamma/8\pi^2) M_{\text{Pl}}^2 H^2$, so $\Omega_{\text{eff}} = a_\gamma/(8\pi^2)$. The same hierarchy appears on three levels — $O(1)$ (the dimensionless ratio), 10^{122} (energy density, M_{Pl}^2/H^2 , equal to the horizon area A_2 in Planck units), and 10^{244} (four-volume, $(M_{\text{Pl}}/H)^4 = V_4$) — and the 10^{240} failure mode of the dynamical Mechanism 2 (Section 14) is exactly this V_4 factor, not an artifact. P5 is what bridges the dimensionless ratio to the observable density. It is supported by the 't Hooft–Susskind area bound (which N_{eff} saturates up to a factor π) and by the dimensional identity $V_4 = A_2^2$ (algebraically trivial, a consistency check only). The substantive content is not the count but its *coherence*: the modes must add linearly ($\rho_{\text{eff}} = N\rho$), not in random phase ($\sqrt{N}\rho$, which would give $\Omega \sim 10^{-64}$, catastrophically wrong). Three structural mechanisms enforce coherence: (K1) Reeh–Schlieder vacuum entanglement makes the local trace expectation $\langle T^\mu{}_\mu \rangle = a_\gamma E_4$ pointwise deterministic rather than stochastic; (K2) the integrated anomaly is topologically fixed, $\int_{S^4} E_4 = 64\pi^2$, independent of the mode number, so the modes must sum linearly to reproduce it; (K3) the KP global constraint $\Lambda V_4 = \int \sqrt{g} \langle T^\mu{}_\mu \rangle$ synchronizes sub-regions, forbidding out-of-phase oscillation. Each independently excludes the random-phase regime. The *coherence* half can moreover be reduced beyond bare motivation: in Mottola’s effective theory of the conformal anomaly [33, 34], the anomaly vacuum energy is a *condensate* of a 4-form field strength $F = dA$ — the local conformal form of the nonlocal anomaly action — i.e. a single coherent classical amplitude over the horizon volume, and a condensate adds in phase ($\times N$) by construction, the $\sqrt{N}$ regime requiring incoherent modes a condensate does not possess. The infrared-relevance of the same action places the backreaction at the horizon scale rather than the UV, with the value set by horizon boundary conditions [33]; Mottola’s $\Lambda \rightarrow 0$ infrared fixed point is then the sequestering of the bare term (Section 8), while the finite $4a_\gamma$ is the anomaly-residual condensate value — two distinct contributions, consistent. What is *still* not available is a strict Schwinger–Keldysh or BV-master-equation derivation fixing the *exact* $O(1)$ normalization of $N_{\text{eff}} = (M_{\text{Pl}}/H)^2$, and Mottola’s conformal EFT, though established, is not universally accepted. Thus P5 is the single quantitatively decisive element of the framework that is not derived from first principles: it carries the entire 122-order magnitude hierarchy, with its coherence now a *consequence of the anomaly condensate* rather than a bare postulate, leaving the residual open piece as the one $O(1)$ horizon normalization. The Euclidean ratio is theorem-level; the magnitude of the physical curvature density rests on P5.

The normalization Mottola leaves open is fixed by the Q-charge; P5 is the declared founding assumption. Mottola’s mechanism does not, by itself, fix the value: it “depends on macroscopic boundary conditions at the cosmological horizon scale” [33], and the conformal theory — hence a_γ — is unspecified. These are exactly what CSG–KP supplies and a generic anomaly EFT does not. The boundary condition is the Q -curvature charge of the hemisphere, $C_Q = 8\pi^2$ (Chang–Yang), and the conformal theory is the photon ($a_\gamma = 31/180$), so the accumulation is the de Sitter entropy per Q -charge,

$$N_{\text{eff}} = \frac{S_{dS}}{C_Q} = \frac{8\pi^2 (M_{\text{Pl}}/H)^2}{8\pi^2} = \left(\frac{M_{\text{Pl}}}{H} \right)^2, \quad (28)$$

with the $O(1)$ factor *exactly* unity because S_{dS} and C_Q share the same $8\pi^2$ normalization. The finite value Mottola’s $\Lambda \rightarrow 0$ fixed point does not provide is then the anomaly residual

$\Omega_\Lambda = 4a_\gamma$, fixed by a_γ and C_Q . What was an undetermined $O(1)$ in the generic EFT is therefore closed by the framework’s own topology. The residual is not a hidden gap but a single *declared* founding assumption: P5 := ($N_{\text{eff}} = S_{\text{dS}}/C_Q$, coherent accumulation as entropy per Q -charge), as fundamental as the holographic principle itself. Given P5, the framework is internally complete — value, sign, ratio, and dark-sector interaction all follow — and its honesty lies in the explicitness of that one postulate together with a sharp falsification channel (the curvature ratio, DESI/Euclid). This is the ordinary structure of a physical theory, not an outstanding deficit.

P5 is built from established results: the entanglement area law and Jacobson’s equilibrium, exact for the photon. P5 has two parts, and the literature supplies both (the value excepted), so it is the application of recognized results to the photon substrate rather than an isolated postulate. First, its area scaling $N_{\text{eff}} = (M_{\text{Pl}}/H)^2$ is the entanglement area law: the vacuum entanglement entropy of a region scales with its boundary area as $A/4L_p^2$, the Bekenstein–Hawking form [35, 36], and $S_{\text{dS}}/C_Q = (M_{\text{Pl}}/H)^2$. Second, the vacuum-entanglement-to- Λ link is Jacobson’s entanglement equilibrium [35]: the semiclassical Einstein equation with a cosmological constant holds if and only if the vacuum entanglement entropy of small causal diamonds is stationary at fixed volume — and crucially this result is *exact* for first-order variations of the vacuum of *conformal* fields (only modulo a conjecture for nonconformal ones). The photon is conformal and the unique infrared-surviving field (Komargodski–Schwimmer), so Jacobson’s theorem applies to the CSG photon substrate exactly: this lifts K1 (Reeh–Schlieder vacuum entanglement) from a structural motivation to an established theorem for precisely the relevant field type. Moreover the maximally symmetric causal diamond that carries the equilibrium is itself de Sitter space with a cosmological constant [37], i.e. the CSG cap, and the negative diamond temperature it requires matches the Hartle–Hawking weighting e^{-S_E} ; while the coherence ($\times N$, not $\sqrt{N}$) is exactly the strict area law that deriving the Einstein equation demands [38]. What no such programme provides is the *value*: Jacobson leaves Λ an integration constant and Verlinde’s dark-energy scale carries contested assumptions. The value $\Omega_\Lambda = 4a_\gamma$ is fixed by CSG’s a_γ (photon) and C_Q (hemisphere); an assumption-free theorem *with* the value is absent across the literature, a gap of the field rather than of CSG specifically. P5 thus stands on the area law and Jacobson’s equilibrium for the conformal photon, with the value as CSG’s own input.

P5 as Jacobson’s theorem with two premises derived and the third triangulated. Jacobson’s derivation rests on three inputs, and the decisive observation is that CSG’s anomaly and infrared structure supply precisely the two that are contested in the literature. *(i)* A universal area density of vacuum entanglement entropy: by Solodukhin and Casini–Huerta the universal four-dimensional entanglement entropy is the conformal anomaly, its type-A part fixed by the topology of the entangling surface (the extrinsic-curvature contribution drops for the spherical de Sitter hemisphere), so the universal coefficient is a_γ , rendered finite by the topological charge C_Q rather than by a cutoff. The known Maxwell subtlety — the naive free logarithmic coefficient differs from the anomaly because of edge modes — is resolved through the four-sphere partition function, which is exactly the object $\zeta(0; \text{Maxwell}; S^4) = -4a_\gamma$ that CSG uses. *(ii)* A conjecture that the nonconformal variation matches the conformal form, which Casini, Galante and Myers found to conflict for relevant operators of low conformal dimension: this case is voided in CSG because the Komargodski–Schwimmer a -theorem leaves only the conformal photon in the infrared, so the problematic nonconformal sector never enters. The remaining input, *(iii)* that the vacuum is an entanglement equilibrium (the entropy stationary at fixed volume), is the one CSG does not prove; it is concretized as the cap-saddle stationarity $d\Gamma/d\theta = 0$ at the de Sitter diamond. This residual premise is not proved but *triangulated* by convergent results of the full pipeline: the cap saddle $\delta\theta^*$ is a genuine action stationary point; the stationary action equals the anomaly invariant $\int_{S^4} \langle T \rangle = 4a_\gamma$; the Riegert scale flow $d\Gamma/d\sigma = 4a_\gamma$ reaches the same invariant by an independent route; the entropy whose stationarity is at issue is $N_{\text{eff}} = S_{\text{dS}}/C_Q$; and the KP vol-

ume constraint $\Lambda V_4 = \int \sqrt{g} \langle T \rangle$ is the Jacobson–Visser first law of the diamond. All five identify the cap saddle with the stationarity of the anomaly action and entropy at the de Sitter diamond, which Jacobson–Visser reformulate as the entanglement equilibrium [37]; the numerical convergence of the action invariant and the Riegert flow on the same $4a_\gamma$ by independent computations is the sharpest leg of the triangulation. P5 is thereby lifted from an isolated postulate to Jacobson’s entanglement-equilibrium theorem for the conformal photon, with premises (i) and (ii) derived from CSG and (iii) triangulated. This is a qualitative reduction, not an assumption-free closure: premise (iii) remains a physical equilibrium hypothesis — Jacobson’s own, and broadly accepted — the Maxwell edge-mode coefficient is still under debate, and Jacobson’s equivalence is established at first order in the variations.

The honest boundary, drawn smaller: one equilibrium premise. Once the pipeline is convergent, two reductions follow. First, the Hartle–Hawking versus Vilenkin branch choice ceases to be an independent postulate. The entanglement-equilibrium premise is defined about the maximally symmetric vacuum, which for the conformal field is the Bunch–Davies state; by Gibbons–Hawking this state restricted to the static patch is a thermal equilibrium, and the Hartle–Hawking vacuum is stationary, $\rho \sim e^{-\beta H}$, whereas Vilenkin’s tunnelling state is an out-of-equilibrium initial condition. Given the equilibrium premise the branch is therefore forced to Hartle–Hawking, with the caveat that in the bare wave-function formalism the sign is not fixed absolutely, so the forcing holds only relative to the premise. Second, the seemingly separate assumptions — A5 (the anomaly action $4a_\gamma$ is the curvature), premise (iii) (the vacuum is an entanglement equilibrium), and P5 ($N_{\text{eff}} = S_{dS}/C_Q$) — are one statement in three forms, since $\Omega_\Lambda = 4a_\gamma$ is exactly equivalent to the normalization $N_{\text{eff}} = S_{dS}/C_Q$ and Jacobson identifies the equilibrium stationarity with that same action invariant. Counting them separately triple-counts a single premise. The exhaustion no-go reinforces this: no dynamical route yields $4a_\gamma$ — every dynamics gives a different number — and only the action invariant $\int_{S^4} \langle T \rangle = 4a_\gamma$ does, so the value cannot be dynamical and the governing principle must be one of stationarity, which is precisely the equilibrium premise. The ledger thus reduces from two postulates to one: a single equilibrium premise — Jacobson’s own, broadly accepted, and forced in form by the no-go — from which the branch follows and from which, with premises (i) and (ii) supplied by CSG, the value follows. The residual is the strict functional identity between the cap-saddle stationarity and Jacobson’s entanglement stationarity, triangulated by the convergent pipeline but not proved. This is the smallest honest boundary the framework reaches: one equilibrium postulate and one triangulated identity, not a first-principles closure.

Reverse anchoring: CSG–KP supplies the number established frameworks must postulate. P5 and the absolute identification do not stand alone; they connect to established holographic dark-energy programmes by a reverse bootstrap. Those programmes fix the dark-energy density through a holographic relation containing one dimensionless $O(1)$ number they must *fit or postulate*; CSG–KP supplies that number from a_γ , and in return the programme supplies the energy-*density* form that the no-go of Section 16.1 showed the anomaly cannot produce dynamically. Concretely, holographic dark energy [29, 30] posits the genuine density $\rho_{\text{DE}} = 3c^2 M_{\text{Pl}}^2 / L^2$ with $c \sim 0.8$ fitted; taking the Hubble-scale cutoff $L = 1/H$ and the CSG–KP value $c^2 = 4a_\gamma$ gives

$$\rho_{\text{DE}} = 3(4a_\gamma) M_{\text{Pl}}^2 H^2 \implies \Omega_\Lambda = 4a_\gamma, \quad (29)$$

so $4a_\gamma$ appears as a real density fraction, not merely as an action invariant, and the predicted coefficient $c = \sqrt{4a_\gamma} = 0.830$ matches the holographic fit. The same number recurs as the holographic-equipartition factor: the bridge ratio $\rho_\Lambda / \rho_{\text{anom}} = 32\pi^2 (M_{\text{Pl}}/H)^2 = 4S_{dS}$ is the de Sitter entropy times $\chi(S^4)^2 = 4$, with the surface form $N_{\text{sur}} = N_{\text{bulk}}$ supplied by emergent gravity [39]. *Honest caveats.* The Hubble-cutoff HDE has $w_{\text{DE}} = 0$ classically [31]; recovering $w_{\text{eff}} = -1$ together with a constant ρ_{DE}/ρ_m (the coincidence) requires a dark-sector interaction

[32]. That interaction is not a free addition, however: the next paragraph shows it is *forced* by the Bianchi identity once the vacuum runs, with its strength set by a_γ . And CosMIn [39] does not reverse-extract: it fixes the *absolute* scale ΛL_P^2 through H_0 , whereas $4a_\gamma$ is the *fraction*; the two are complementary, not reducible. The net gain is real but bounded: $4a_\gamma$ now carries a density realization and is anchored independently in the HDE coefficient, the equipartition factor, and the anomaly action — no single one proving it, but together removing its isolation. The residual open step is no longer “action or density” but the choice of the Hubble cutoff and the exact strength of the (Bianchi-forced) interaction.

The dark-sector interaction is forced, not free: running vacuum from the anomaly.

The interaction the previous paragraph needs is not an extra ingredient but a consequence of energy conservation. If the vacuum energy runs, $\rho_{\text{vac}} = \rho_{\text{vac}}(H)$, the total Bianchi identity $\nabla_\mu T_{\text{tot}}^{\mu\nu} = 0$ *forces* an exchange with matter that cannot be switched off; this is the running-vacuum mechanism [40, 41]. The forced continuity equation is

$$\dot{\rho}_m + 3H\rho_m = +3\nu H\rho_m \implies \rho_m \propto a^{-3(1-\nu)}, \quad (30)$$

with a single dimensionless strength ν equal to the β -function coefficient of the running cosmological constant — a QFT quantity, $\nu_{\text{eff}} = \nu_s + \nu_f + \nu_v$, to which vector bosons contribute even in the conformal limit ($\nu_s = 0$ but $\nu_v \neq 0$) [41]. In CSG–KP the only infrared-surviving anomaly source is the photon, so $\nu = \nu_v(\text{photon})$, and the natural anomaly magnitude is

$$\nu = \frac{a_\gamma}{8\pi^2} = \frac{31}{1440\pi^2} \approx 2.18 \times 10^{-3}, \quad (31)$$

which lies inside the running-vacuum data-fit band $|\nu| \sim 10^{-3}\text{--}10^{-2}$ *and* is numerically identical to the curvature ratio Ω_K/Ω_Λ of Eq. (1). The same anomaly number thus drives both the curvature prediction and the dark-sector exchange rate, so the coincidence resolution introduces *no* new parameter, and $w_{\text{eff}} = -1$ is recovered from the running vacuum rather than from a free Hubble-cutoff fluid, bypassing the Hsu obstruction. *Two honest caveats remain.* First, $\nu = a_\gamma/(8\pi^2)$ is the natural choice with the correct magnitude, but the exact ν_v for a massless vector field via adiabatic regularization has not been computed here — the magnitude and the consistency stand, the sharp derivation does not. Second, generic phenomenological Hubble-cutoff interactions are excluded by CMB/LSS data; running vacuum is a distinct, field-theoretically grounded class, only partly tested, so the specific CSG–KP value of ν must still be confronted with structure-formation data. This reduces point B from “needs a new interaction parameter” to “needs the anomaly number $a_\gamma/(8\pi^2)$, already present as the curvature ratio” — progress, not closure. *A third caveat must be stated plainly:* this mechanism yields $w_{\text{eff}} = -1$ as the CSG–KP *prediction*, but DESI’s $w_0 w_a$ data prefer a *dynamical* w at $\sim 4\sigma$ (if real), a deviation of order 0.25–0.6 that an anomaly number $\nu \sim 2 \times 10^{-3}$ is two orders of magnitude too small to produce. The running-vacuum term keeps $w_{\text{eff}} = -1$ self-consistently, but it does *not* explain — and cannot absorb — the DESI w -preference; that remains the framework’s sharpest open empirical tension, stated as such in the robustness analysis below (Section 25).

The two open steps are one: a massless source forces the accumulation. Pressing the interaction calculation harder settles both residual steps at once, by showing they are the same assumption. The running-vacuum $O(H^2)$ term that acts today, $\delta\rho_{\text{vac}} \sim \nu_{\text{eff}} M_{\text{Pl}}^2 H^2$, originates in field *masses* (terms $\sim m_i^2/M_{\text{Pl}}^2$; [41]), so a massless photon gives *no* $O(H^2)$ contribution, $\nu_v(O(H^2)) = 0$. The photon enters only at $O(H^4)$ — the conformal-anomaly density $\rho_{\text{anom}} = 3a_\gamma H^4/8\pi^2$ — which today, with $H^2/M_{\text{Pl}}^2 \sim 10^{-122}$, is 122 orders too small to act as the $O(H^2)$ interaction. The naive extraction of ν from a massless field thus fails. It is *the same* P5 accumulation $N_{\text{eff}} = (M_{\text{Pl}}/H)^2$ that rescues it: lifting the $O(H^4)$ term gives the Friedmann

fraction

$$\frac{N_{\text{eff}} \rho_{\text{anom}}}{3M_{\text{Pl}}^2 H^2} = \frac{a_\gamma}{8\pi^2} = \frac{\Omega_K}{\Omega_\Lambda}, \quad (32)$$

exactly the curvature ratio, and the Hubble cutoff $L = 1/H$ of the density bridge is that *same* de Sitter horizon scale, $N_{\text{eff}} = (M_{\text{Pl}}/H)^2$ = the horizon area in Planck units. Hence the absolute value (which needed the cutoff) and the interaction strength (which needed the lift) are not two independent assumptions but one: the coherent holographic accumulation at the de Sitter horizon. The coherence motivation transfers intact — the sharpest piece, K2, anchors both to the same topological invariant, since the value uses $\int_{S^4} E_4 = 64\pi^2$ and the interaction’s vacuum-running coefficient is fixed by the same flux through the Riegert flow $d\Gamma/d\sigma = (a_\gamma/16\pi^2) \int_{S^4} E_4 = 4a_\gamma$. The net effect is a reduction of the open surface from two seemingly separate conjectures to one — coherent saturation of the holographic bound, $N_{\text{eff}} = S_{dS}/(8\pi^2)$, as fundamental as the holographic principle itself and still lacking a strict Schwinger–Keldysh or BV derivation. A sharper localization of what is open, not a closure.

Observer-timelessness: the ratio is epoch-invariant, the absolute scale need not be. The $N_{\text{eff}} = M_{\text{Pl}}^2/H^2$ compensation has a consequence that reframes the coincidence “why now”. Because $N_{\text{eff}} \sim H^{-2}$ and $\rho_{\text{anom}} \sim H^4$, the factor H cancels *entirely*: $\Omega_{\text{eff}} = a_\gamma/(8\pi^2)$ holds at every epoch, independent of H . The photon substrate is lightlike and carries no proper time, so the prediction — the ratio Ω_K/Ω_Λ — is observer- and epoch-invariant, while the absolute Ω_Λ carries the scalaron position φ_0 , i.e. “where/when the observer is”, and is in this reading observer-dependent rather than predicted. The “why now” is then dissolved as observer-selection on the absolute scale, the timeless ratio needing no tuning. This is the honest counterpart to the geometric ladder of Section 16.1: the ladder is the sharper but *conjectural* alternative, in which even the absolute $\Omega_\Lambda = 4a_\gamma$ is topological (full S^4); the observer-timelessness reading is the conservative one, in which only the ratio is fixed. The two are in tension — the ladder would make Ω_Λ epoch-marked but topologically determined — and which holds depends on the open action-identification of Section 16.1.

20 Higher-order corrections

20.1 The two-loop coefficient $c_1 = 2\zeta(3)$

The Coleman–Weinberg expansion of the scalaron potential carries a two-loop coefficient $c_1 = 2\zeta(3)$, taken as a *literature-supported* input from Mottola–Vaulin [21]. The primitive period $6\zeta(3)$ of the underlying sunset master integral is rigorous; the normalization $6\zeta(3) \rightarrow 2\zeta(3)$ is the Mottola–Vaulin photon-mediated value. A naive σ^3 sunset gives $\frac{2}{9}\zeta(3)$, the photon-polarization-enhanced version $\frac{8}{9}\zeta(3)$; the remaining factor (a Wess–Zumino-cocycle contribution) is supplied by the literature value. This is a deliberate and clearly bounded reliance on a peer-reviewed result; Section 21 (M1- c_1) confirms the literature input rather than revising it.

20.2 The Coleman–Weinberg amplitude

At the Hubble scale all massive fields are infrared-decoupled, leaving the photon with effective coupling $g_{\text{eff}}^2 = a_\gamma$. Naive-dimensional-analysis loop counting with two Wess–Zumino anomaly insertions over two four-dimensional loop integrations gives the M5 amplitude

$$A_{\text{M5}} = \frac{a_\gamma^2}{(16\pi^2)^2} \approx 1.19 \times 10^{-6}, \quad (33)$$

and including the two-loop coefficient, $c_1 A_{\text{M5}} = 2\zeta(3) A_{\text{M5}} \approx 2.86 \times 10^{-6}$. The $(16\pi^2)^2$ structure is fixed by the two-loop counting.

20.3 Convergence of the loop series

The three-loop sunset master integral $T_{3SS} = -3\zeta(5) + 5\zeta(3)^2 - 2\zeta(3)$ gives (symmetry factor $1/144$, Antoniadis–Mottola vertex $2/3$)

$$c_2 = -\frac{1}{108}\zeta(5) - \frac{1}{162}\zeta(3) + \frac{5}{324}\zeta(3)^2 \approx 5.3 \times 10^{-3}, \quad |c_2(c_1 A_{M5})^2| \approx 4.3 \times 10^{-14}, \quad (34)$$

where the relevant amplitude is the two-loop value $c_1 A_{M5} \approx 2.86 \times 10^{-6}$ of Eq. (33), not A_{M5} alone. Each loop order is suppressed by $a_\gamma^2/(16\pi^2)^2 \approx 1.2 \times 10^{-6}$; the three-loop contribution is roughly nine orders of magnitude below DESI DR3 sensitivity. This is a *proven* bound (FORM-verified master integrals, Section 21 M1-3L).

21 Independent computational verification (M1)

The framework’s key quantitative claims have been independently recomputed with FORM 5.0.0, SymPy (exact symbolic arithmetic), and SciPy (numerical eigenvalue solvers). No result contradicts the main text; the open items in particular are *corroborated*, not revised. The standalone code accompanying this work reproduces each of the following.

M1-Q. The Q -charge $\int_{D^4} Q_4 = 8\pi^2$ is verified by exact symbolic arithmetic ($Q_4 = 6$ on the unit S^4 ; $\int_0^{\pi/2} \sin^3 = 2/3$; $6 \cdot 2\pi^2 \cdot 2/3 = 8\pi^2$), matching Lemma 1.

M1- ζ . The DtN identity is verified in both bookkeepings of Section 5.2; multi-gauge cross-checks (Lorenz and Coulomb) reproduce the hemisphere value and return the closed value $-31/45$.

M1-3L. The three-loop Coleman–Weinberg bound $|c_2(c_1 A_{M5})^2| \approx 4.3 \times 10^{-14}$ is reproduced from the FORM master integrals.

M1-H. The cap-saddle stability is reconfirmed: shooting-method eigenvalues in the scalar, 1-form, and TT sectors are all positive on the deformed hemisphere (e.g. scalar $10.0 \rightarrow 8.16$, 1-form $12.71 \rightarrow 10.67$, TT $20.0 \rightarrow 17.27$ for the lowest modes), with the single negative lapse direction unchanged.

M1- Λ . The trace-zero identity $\Lambda^* = \frac{1}{4}\langle R \rangle \Leftrightarrow \langle T \rangle = 0$ is verified symbolically; the closed-history requirement (finite V_4 for $k = +1$, divergent for $k = 0, -1$) and the unique negative Coleman–Weinberg vacuum at $\varphi_{\min} = v s e^{11/6}$ are reproduced.

M1-FRG. The functional-renormalization-group analysis corroborates the assessment that Interpretation A (the RG-attractor route for A2) is insufficient — the dimensionless ratio overshoots the target by a factor ~ 13 with no attractor. A2 is instead *numerically established* in its physically relevant form, Interpretation B: the iterative fixed point of the KP self-consistency map (M5 Banach contraction $L \approx 0.18$, sign $G' < 0$ analytic, $|G'| < 2$ bound numerical, Section 10). The retracted RG reading and the iterative fixed point are not in conflict — they concern flow in scale versus iteration at fixed cosmological scale — and the redshift-constancy of the central ratio is in any case topological, independent of either.

M1- c_1 . The literature input $c_1 = 2\zeta(3)$ is confirmed, not revised.

22 Independent computational verification (M1), in detail

This section expands Section 21. The recomputation used FORM 5.0.0, SymPy (exact) and SciPy (numerical); no result contradicts the main text, and the FRG and c_1 items corroborate the existing treatment rather than revising it.

- *Q-charge:* SymPy reproduces $Q_4(S^4) = \frac{1}{6}(144 - 108) = 6$, $\int_0^{\pi/2} \sin^3 = \frac{2}{3}$, $C_Q = 6 \cdot 2\pi^2 \cdot \frac{2}{3} = 8\pi^2$ exactly; the cap-shift is a wave-function argument position, so L_0 carries no $\delta\theta^*$ correction.

- $\zeta(0; \text{DtN}) = -1$: confirmed in multiple gauges; the Esposito decomposition $\zeta_{\text{PDF}} = -77/180$, $\zeta_{R_1} = -1/20$, $\zeta_{\text{GM}} + \zeta_{\text{gh}} = -11/30$ sums to $-38/45$, with the BFK route rigorous and self-contained.
- *Three-loop bound (FORM)*: $T_{3\text{SS}} = -3\zeta(5) + 5\zeta(3)^2 - 2\zeta(3)$, $c_2 \approx 5.3 \times 10^{-3}$, $|c_2(c_1 A_{\text{M5}})^2| \approx 4.3 \times 10^{-14}$ (the amplitude is the two-loop value $c_1 A_{\text{M5}} \approx 2.86 \times 10^{-6}$).
- *Cap-saddle stability* (analytic, by domain monotonicity; confirmed by shooting): scalar $10.0 \rightarrow 8.16, \dots, 54.0 \rightarrow 48.06$; transverse 1-form (Weitzenböck-distinct) $12.71 \rightarrow 10.67, \dots, 56.25 \rightarrow 50.21$; TT $20.0 \rightarrow 17.3, \dots, 59.7 \rightarrow 53.5$; all positive (each cap value lies strictly above the closed- S^4 limit $\geq 4, 6, 12$), lapse mode Morse index 1 unchanged.
- *FRG corroboration*: an independent Reuter–Saueressig flow finds a clean UV fixed point but the ratio $R = \kappa/\lambda$ overshoots $a_\gamma/(8\pi^2)$ by a factor 13 at $t = -1$, with no attractor — confirming that A2 as an RG attractor fails. A2 is instead numerically established via Interpretation B (the iterative fixed point of the KP self-consistency map, M5 Banach contraction; sign $G' < 0$ analytic, $|G'| < 2$ numerical); the *topological* z -independence of the ratio (the framework’s actual argument) is untouched by either reading.
- $c_1 = 2\zeta(3)$: FORM reproduces the Mottola–Vaulin normalization; the primitive period $6\zeta(3)$ is rigorous and the factor $6\zeta(3) \rightarrow 2\zeta(3)$ is the literature input.

23 Observational status and forecasts

23.1 Current pull

With the ladder-consistent budget $\Omega_m = 0.3096$ and the open sign, the present-epoch prediction is $\Omega_K^{\text{obs}} = +0.00150$ (L0) or $+0.00143$ (L1). The most recent curvature-sensitive analysis of DESI DR2 + Planck [23] reports $\Omega_K^{\text{obs}} = +0.0023 \pm 0.0011$, a mild (2.1σ) preference for an open universe. The framework pull is $(0.0023 - 0.00150)/0.0011 \approx 0.72\sigma$ (L0), 0.79σ (L1) — fully consistent, and on the open side. A preliminary MCMC run imposing the $\mathcal{L}$ -constraint over Planck plik + DESI DR2 returns $\Omega_K^{\text{obs}} = +0.00152$ with $\Delta\chi^2 = -10.77$ versus flat ΛCDM and a Hubble constant $H_0 = 68.6$ shifted modestly toward the local-distance-ladder value. We flag the chains, priors, and likelihood configuration as not yet released in publication-grade form, and we therefore do *not* treat this run as evidence for the framework; a full, reproducible Cobaya analysis is forthcoming.

23.2 Forecasts and discrimination

Survey (epoch)	σ_K	Detection of central value	L0-vs-L1 discrimination
DESI DR3 (~ 2027)	$\sim 3 \times 10^{-4}$	$\sim 5\sigma$	$\sim 0.24\sigma$
Euclid DR1	$\sim 1.5 \times 10^{-4}$	—	$\sim 0.48\sigma$
Euclid + CMB-S4 (2030+)	$\sim 5 \times 10^{-5}$	—	$\sim 1.4\sigma$

Table 3: Forecast sensitivities. DESI DR3 will test the central relation at $\sim 5\sigma$; the L0/L1 refinement requires the combined Euclid + CMB-S4 sensitivity.

24 Numerical MCMC details

Preliminary internal results: chains, priors, and the likelihood configuration are not yet released in publication-grade form. The numbers below are indicative only and are not used to support the framework’s claims; a complete, reproducible Cobaya chain set will be released separately.

A Planck plik + DESI DR2 BAO chain was run with Cobaya (clik likelihoods), eight walkers, 5×10^5 steps. The constraint $\mathcal{L} = 8\pi^2/a_\gamma = 458.4$ is imposed as a derived relation $\Omega_\Lambda = \mathcal{L} |\Omega_K|$ over a free Ω_K , with $\Omega_m + \Omega_\Lambda + \Omega_K = 1$. Sampled: $\Omega_b h^2$, $\Omega_c h^2$, θ_{MC} , τ , n_s , $\ln(10^{10} A_s)$, Ω_K . Relative to flat Λ CDM the chain achieves $\Delta\chi^2_{\text{CMB}} = -6.4$ and $\Delta\chi^2_{\text{BAO}} = -4.4$ ($\Delta\chi^2_{\text{tot}} = -10.77$), with $\Omega_K = +0.00152$ and $H_0 = 68.616$, modestly shifted toward the local-distance-ladder value. Two independent likelihood implementations agree within $\Delta\chi^2 < 1$.

25 Consolidated status and falsifiability

25.1 The epistemic ledger

Table 4: Consolidated status of the framework’s components.

Component	Status	Basis
Q -charge $\int_{D^4} Q_4 = 8\pi^2$	proven	Chang–Yang/Branson; M1-Q
Spectral results R1–R12	proven	spectral computation; [11]
$\zeta(0; \text{Maxwell}; S^4) = -31/45$ (R1)	proven	two derivations
BFK identity (both bookkeepings)	proven	[10]; M1- ζ
Hemisphere quotient $a_\gamma/(8\pi^2)$ (L0)	proven	topological identity (Sec. 11)
Cap-saddle $\delta\theta^*_{\text{exact}} = 0.16391$	proven	exact saddle Eq. (19)
Cap-saddle stability (all sectors)	proven (analytic)	domain monotonicity + closed- S^4 spectrum; M1-H
Boundary refinement (BFK odd-function)	proven	rigorous symmetry (Sec. 12.3)
Uniqueness of the invariant	proven (within criteria)	Paneitz zero-mode + (i)–(iv); full minimality open
KP sequestering (magnitude problem)	derived within CSG	Built-in via Stückelberg compensator (proof sketch); full all-orders rigor pending
KP self-consistency (value, structural)	numerically established	Proposition 2 (sign $G' < 0$ analytic, $ G' < 2$ bound numerical); M1- Λ
Three-loop bound $\approx 4.3 \times 10^{-14}$	proven	FORM; M1-3L
$a_\gamma = 31/180$	derived	Gilkey a_4 on S^4 (Sec. 3); scalar check 1/360
$c_1 = 2\zeta(3)$	literature; cancels from ratio	Mottola–Vaulin; convergence only, not in prediction
Absolute Ω_Λ (KP volume)	universality structure only; scale open	Λ^* universal in φ_0/Ω_m (proven) but amplitude-set (≈ -0.10 for A_{CW} , not -0.69); φ_0 free; the $\langle R \rangle/4$ sharpness proxy was a floored-geometry artefact (retracted)

Component	Status	Basis
Budget ladder $\Omega_\Lambda = 4a_\gamma$, $\Omega_K = a_\gamma^2/2\pi^2$	conjecture (sharpened)	ladder algebra exact ($\Omega_K \cdot 32\pi^2 = \Omega_\Lambda^2$, $32\pi^2 = \int_{D^4} E_4$), Planck match $< 1\sigma$; lift prefactor now <i>computed</i> = A/G (Bekenstein–Hawking horizon dof = $4S_{dS}$), residual = CKN satu- ration P5 (named, not proven); Sec. 16.1
P5 localization (in-in)	localized, not proven	Schwinger–Keldysh: anomaly Adler–Bardeen exact; lift = Cohen– Kaplan–Nelson bound; Schwarzschild part fol- lows from gravity (fixes 10^{-122}), saturation = P5 $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ scale- invariant (H, M_{Pl} cancel) $\Rightarrow$ Euclid-falsifiable core needs no P5; absolute Ω_Λ alone carries the satura- tion
Scope (ratio vs absolute)	ratio holography-free	Lombriser 2019; assumes $y(t_0) = \frac{1}{2}$
$\Omega_\Lambda = 0.704$ (halo-averaged)	derived, conditional	apparent-horizon hy- persurface forced ($rH=1$); form not forced (KP/thermo fail, closures spread 0.49–0.96) $\rightarrow O(1)$ band
Absolute Ω_Λ (causal closure)	geometric, prior-free	Cauchy + Chern–Weil + APS; $ \epsilon_\partial \leq 1.5\%$
A5a (topological ratio $a_\gamma/8\pi^2$)	theorem	saddle-point, conditional on B1 (established, re- duced)
A5b (Wick-rotation kinematics)	derived	the only genuinely chosen element; data-anchored
A5c (reference-state choice)	specification	contraction $L = 1 + G' < 1$ requires $-2 < G' < 0$: sign $G' < 0$ analytic, $ G' < 2$ bound numerical
A2 (IR fixed point)	numerically established	not required; topology re- places it (M1-FRG)
A2 as an RG-flow attractor	retracted	unique cap saddle $\delta\theta^* > 0$ for $a_\gamma > 0$; weighting- independent
Curvature sign $\Omega_K > 0$ (open)	forced	conformal mode rotated; transverse modes positive (B1')
Euclidean fluctuation stability	resolved (GHP)	

Component	Status	Basis
Euclidean vs Lorentzian definition	open (interpretational)	FLT debate; not CSG-KP-specific
B1 (cap dominance, reduced action)	established (analytic)	$n_{\text{complex}} = 0$; no Stokes line; gap $> 7 \times 10^5$
B1' (cap + scalar embedding)	complete (analytic)	degree-8 polynomial reduction, both φ -branches; Picard-Lefschetz $\Rightarrow n_{\text{complex}} = 0$
Full no-boundary path integral	assembled (semiclassical)	1-loop det $= -4a_\gamma$; Type-A Adler-Bardeen exact
All-orders stability of $R = a_\gamma/(8\pi^2)$	proven (within CSG)	no graviton loops; ratio of $\hbar$ -indep. invariants; abs. φ^4 Borel decoupled

The observable prediction reduces, after this decomposition, to a single genuinely chosen element: the reference-state specification A5c (the canonical Λ -dominated present epoch), anchored by current cosmological data. Its Euclidean half A5a is a theorem (Cauchy + Chern–Weil + APS) and the Wick-rotation kinematics A5b are derived in saddle-point given B1; the constancy of the ratio in redshift is *topological* (Komargodski–Schwimmer invariance of an integrated anomaly), so the retracted RG-attractor reading of A2 does not bear on it — and A2 itself is *numerically established* in its physically relevant form, the iterative fixed point of the KP self-consistency map (M5 Banach contraction; sign $G' < 0$ analytic, $|G'| < 2$ bound numerical). The cap-saddle dominance B1 is analytically established on the reduced cap-action, and the two-dimensional cap-plus-scalar embedding $B1'$ is complete ($n_{\text{complex}} = 0$ by polynomial reduction, Section 18); the embedding into the full infinite-dimensional no-boundary path integral over all inhomogeneous matter and metric modes is assembled semiclassically and is one-loop exact for the Type-A prediction (Section 18); the prediction is all-orders stable as a ratio of $\hbar$ -independent invariants (no graviton loops in CSG, Adler–Bardeen-exact Type-A, topological $8\pi^2$; Section 18); the one residue is the absolute Borel summability of the φ^4 matter series, which cancels from the ratio and does not bear on the prediction.

25.2 What the framework resolves

The three faces of the cosmological constant problem are addressed as follows. The *magnitude* is handled in two stages: the bare radiative vacuum energy is removed by sequestering (derived within CSG, Section 8), while the magnitude of the anomaly-induced curvature density rests on the holographic accumulation postulate P5 (Section 19), which carries the 122-order hierarchy and is not derived from first principles. The *value* is fixed structurally by Proposition 2: Λ is not free, but the unique fixed point of the averaging map; its absolute scale is a calibration at the homogeneous level. The *coincidence* is addressed by halo-weighted averaging, which returns $\Omega_\Lambda = 0.704$ conditional on the uniform-prior assumption. The *ratio* (1) — parameter-free given the identification A5, its topological core A5a a theorem — links the residual cosmological constant to the spatial curvature and supplies a single sharp test.

Two levels of claim. The results separate onto two levels; conflating them is the main source of overstatement. *The dimensionless prediction* — the ratio $\Omega_K/\Omega_\Lambda = a_\gamma/(8\pi^2)$ and the open sign $\Omega_K > 0$ — is theorem-level up to one data-anchored convention. Each ingredient is proven or derived: $a_\gamma = 31/180$ (Gilkey a_4), $C_Q = 8\pi^2$ (Chang–Yang), the hemisphere quotient, the cap saddle and its *analytic* stability (domain monotonicity), the contraction A2 ($L = |1 + G'| < 1$; sign $G' < 0$ analytic, $|G'|$ bound numerical), the dominance B1 and the complete embedding $B1'$,

the topological half A5a (theorem) and the kinematic half A5b (derived given B1); uniqueness holds within criteria (i)–(iv). The only chosen element is the reference epoch A5c, fixing “today” for a quantity scaling as $(1+z)^2$ — a calibration convention, not a postulate.

The absolute magnitude — why the curvature density has its observed size and why $\Omega_\Lambda \approx 0.685$ — is a logically separate question the dimensionless prediction does not answer. Two former ingredients have moved off the open list: sequestering is *derived within CSG* (the Built-in Sequestering proposition, Section 8: the Weyl R^2 term supplies the Stückelberg compensator realizing the KP constraint — a proof sketch, full all-orders rigor pending), and $c_1 = 2\zeta(3)$ *cancels from the ratio*. What remains not derived from first principles is then narrower: the holographic accumulation P5 (carrying the 122-order hierarchy, its coherence enforced by K1–K3 but not strictly derived, Section 19); and the absolute value Ω_Λ itself. The latter is subtler than a flat “not derivable”: the self-consistency route yields a *universal* fixed point Λ^* (independent of φ_0 and Ω_m) but only amplitude-set (≈ -0.10 for the CW amplitude, not the -0.69 once read off a $\langle R \rangle/4$ proxy that proves a floored-geometry artefact), so φ_0 — hence $\Omega_\Lambda = (V(\varphi_0) - \Lambda^*)/3$ — stays free. The *anthropic* measure problem is categorical but *avoidable*: a causal variational closure needs no measure, and on it the boundary hypersurface is *forced* (the apparent horizon, $\theta = 0$, $rH = 1$). What stays open is then neither measure nor boundary but the closure *functional*: KP stationarity over the cone has no fixed point, apparent-horizon thermodynamics give an identity or $\Omega_\Lambda = 1$, and bare dimensional closures spread across 0.49–0.96 (the 0.725 choice not distinguished). The causal route thus pins the hypersurface and an $O(1)$ range, not a sharp value. The framework therefore delivers a sharp parameter-free *relation*, a universal but amplitude-set fixed point with the absolute scale open, a forced apparent-horizon hypersurface with an $O(1)$ range for Ω_Λ , and — conjecturally — the budget ladder ($\Omega_\Lambda = 4a_\gamma$, matching Planck) awaiting one identification step; but *not* a first-principles value for the dark-energy density, pending either a derived closure functional or that ladder bridge.

25.3 Falsification criteria

1. Ω_K outside $\pm 3\sigma$ of $a_\gamma \Omega_\Lambda / (8\pi^2)$ refutes the central relation.
2. A robust detection of a *closed* universe ($\Omega_K < 0$) refutes the sign prediction.
3. The constant object is the *action-parameter* ratio $\mathcal{R} = 3|K|/\Lambda$; the density-parameter ratio redshifts as $\Omega_K(z)/\Omega_\Lambda(z) = \mathcal{R}/a^2$ and is therefore *not* epoch-independent. The sharp test is the present-epoch value $\Omega_K^{(0)}/\Omega_\Lambda^{(0)} = \mathcal{R}$ (extracted as a derived parameter with $a_0 = 1$): a measured $\Omega_K^{(0)}/\Omega_\Lambda^{(0)}$ outside $\pm 3\sigma$ of $a_\gamma/(8\pi^2)$ refutes the relation.

DESI DR3 will test the first and third within reach; the present $\geq 2\sigma$ preference for an open universe (DESI BAO + CMB + SNe) already constrains the second. The sign evidence is *not* yet robust across datasets, however: while the BAO-combined fits favour $\Omega_K > 0$, Planck temperature data alone pull toward a *closed* universe ($\Omega_K \approx -0.04$ at ~ 2 – 3σ), an effect entangled with the CMB lensing-amplitude anomaly. The theoretical sign is forced ($a_\gamma > 0$), but its observational confirmation remains data-combination dependent until DR3.

25.4 Robustness: objections forced, cleared, or genuinely open

A falsification attempt raised seven objections. Five are forced by the structure or cleared by results elsewhere in the pipeline; one (DESI w -dynamics) is a genuine open empirical threat we do *not* claim to clear; the seventh is the single postulate isolated above. We collect them so the one unresolved item is stated as such, not glossed over.

1. *The open sign is a free choice.* It is not: the reduced cap action has saddle equation $a_\gamma \cos^4 \delta = \sin \delta$, whose only real root is $\delta\theta^* = 0.164 > 0$; a closed universe would require $a_\gamma < 0$, so the sign $\Omega_K > 0$ is forced by $a_\gamma = 31/180 > 0$.
2. *The factor four in $\Omega_\Lambda = 4a_\gamma$ is numerology.* It is the spectral ratio $\zeta(0; \text{Maxwell}; S^4)/a_\gamma = (-31/45)/(-31/180) = 4$, fixed by the BFK completeness identity $2\zeta(0; D^4) - \zeta(0; \text{DtN}) =$

- $-4a_\gamma$; a_γ is the Gilkey coefficient and $\pi^3 \approx 31$ is flagged as coincidence, not used.
3. *No dynamics produces the value, so it is unfounded.* The no-go is a feature: $\Omega_K/\Omega_\Lambda = a_\gamma/8\pi^2$ is a topological ratio protected by conformal invariance, characterising the geometry at all epochs, not an initial condition propagating through Friedmann evolution. Deriving it dynamically is a category error, and the absence of any dynamical route is consistent with the topological reading rather than evidence against it.
 4. *The Maxwell entanglement coefficient is $-16/45$, not $-31/45$.* The $1/3$ gap is the gauge edge-mode contribution (Casini–Huerta–Magán–Pontello), and the DtN determinant measures it exactly: $\zeta(0; \text{DtN}; \text{Maxwell}; S^3) = 3 = \dim S^3$ counts the boundary polarizations, and through the sphere factorization $Z_{S^4} = Z_{\text{bulk}}/Z_{\text{edge}}$ the DtN determinant is Z_{edge} . The discrepancy is encoded, not missed, and is itself an independent lattice-testable prediction.
 5. *Why only the photon; the massless graviton should contribute.* The graviton is non-conformal (the Lichnerowicz operator’s Riemann term breaks Weyl invariance, so the conformal result does not apply and it drops out for the same reason massive fields do), it is the geometry being determined (its inclusion is circular), and on D^4 only χ survives ($\sigma = 0$). The photon is the unique massless conformal non-gravitational field.
 6. *DESI prefers dynamical dark energy at $> 3\sigma$, refuting $w = -1$.* This is the one objection that is *not* cleared. The hope that DESI’s $w_0 w_a$ preference (obtained under forced $\Omega_K = 0$) is an artefact of the Ω_K – w_0 degeneracy fails quantitatively: opening to $\Omega_K = +1.5 \times 10^{-3}$ shifts the effective w_0 by only $\sim 2 \times 10^{-3}$, whereas DESI requires 0.25 – 0.6 —a factor ~ 100 – 300 too small, and the running-vacuum strength $\nu = a_\gamma/8\pi^2 \approx 2.2 \times 10^{-3}$ is equally small. The framework therefore carries *two* independent tensions with DESI: curvature ($\sim 1.8\sigma$, tolerable) and w ($\sim 4\sigma$ if real), and the second cannot be absorbed by the first. Honest status: either DESI’s w -signal weakens with data, or $w \neq -1$ is real and the constant $\Omega_\Lambda = 4a_\gamma$ requires an extension. This is a genuine open empirical threat, decided by DESI DR3 and Euclid, not by the present analysis.

The honest boundary is therefore exactly *one conceptual postulate* — the anomaly–curvature identification $A5 = (\text{iii}) = P5$ — *together with one sharp empirical threat* ($w \neq -1$), both decided by observation.

Acknowledgement of conditional status

This paper separates what is proven from what is assumed. The Euclidean conformal and topological core — the Q -charge, the twelve spectral results, the BFK identity, the hemisphere quotient, the cap-saddle and its stability, the boundary refinement, the uniqueness statement, and the structural content of the self-consistency theorem — is established at theorem level and independently recomputed (M1). The photon anomaly coefficient $a_\gamma = 31/180$ is derived from the Gilkey heat-kernel coefficient (Section 3); the two-loop normalization $c_1 = 2\zeta(3)$ enters only the convergence bound and cancels from the predicted ratio. Sequestering is derived within CSG (the Built-in Sequestering proposition: the Weyl R^2 term supplies the Stückelberg compensator realizing the KP constraint — a proof sketch, full all-orders rigor pending), not an external add-on. The absolute dark-energy scale is a calibration, becoming a conditional derivation only with the structure-formation assumption $y(t_0) = \frac{1}{2}$. The KP self-consistency fixed point A2 is numerically established (M5 Banach contraction; sign analytic, $|G'|$ bound numerical); cap-saddle dominance B1 is analytically established on the reduced cap-action; and the Lorentzian identification A5 decomposes into a theorem (A5a), a B1-conditional kinematic derivation (A5b), and a single data-anchored choice (A5c). The open curvature sign is forced by $a_\gamma > 0$ via the unique cap saddle. What remains genuinely open is: the holographic accumulation postulate P5 (which carries the 122-order magnitude hierarchy and lacks a Schwinger–Keldysh derivation of $N_{\text{eff}} = M_{\text{Pl}}^2/H^2$); the choice A5c; the absolute Borel summability of the φ^4 matter series (the prediction itself being all-orders stable as a ratio of $\hbar$ -independent invariants, Section 18); and

the Lorentzian fluctuation stability (the Feldbrugge–Lehners–Turok question) — the last of these is where observation, not further algebra, will decide.

26 Code and data availability

A self-contained Python package reproduces every numerical and symbolic statement of this paper. The headline constants and the central prediction are defined once in `csg_kp_core.py` (the single source for $a_\gamma = 31/180$, $C_Q = 8\pi^2$, the matter fraction $\Omega_m = 0.3096$, and the derived Ω_Λ , Ω_K); all other modules import from it. The full verification harness is `run_all.py` (37/37 modules), with the dedicated falsification and audit harnesses `foundations/falsification_suite.py` (16/16 checks) and `foundations/axiom_status_audit.py`. Individual claims map to modules as follows.

- $a_\gamma = 31/180$ from the Gilkey heat-kernel coefficient (independent of the core, to verify it): `foundations/a_gamma_derivation.py`.
- Universality of the Banach fixed point Λ^* and the Coleman–Weinberg iteration: `foundations/cw_banach_iter.py`.
- Coherent spectral accumulation lifting the local anomaly to the curvature scale: `foundations/spectral_coherence.py`.
- The geometric budget ladder $\Omega_\Lambda = 4a_\gamma$ from a_γ and $\chi = 2$: `foundations/budget_ladder.py`.
- The A/G lift prefactor (Bekenstein–Hawking horizon degrees of freedom): `foundations/action_density_bridgman.py`.
- The Schwinger–Keldysh localization of the residual assumption P5 (CKN saturation): `foundations/schwinger_keldysh.py`.
- The scope confinement (conjecture constrains the absolute value, not the ratio): `foundations/ratio_vs_absolute.py`.
- The no-go audit for the absolute $\Omega_\Lambda = 4a_\gamma$ value: `foundations/absolute_value_audit.py`.
- The three residual foundational points (sign, $B1'$, reference epoch): `foundations/close_open_points.py`.

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